

*N m'a faamu**

Boosting Learning without Dividing: Evidence from a Bilingual Education Reform in Mali

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Abstract

Despite promising evidence on the benefits of multilingual education in Sub-Saharan countries, its expansion suffers from a lack of evidence on linguistic transfer from the first language to the second language and a persistent belief that it will boost ethnic identity. This study evaluates the long-term impacts of Mali's 1999 national bilingual education reform, which introduced 11 local languages into the primary school curriculum as languages of instruction. Exploiting spatial variation in policy implementation and using a difference-in-differences approach with commune-level data, the study finds that exposure to bilingual education significantly improves local language literacy by 30% and French literacy by 10%. These gains are especially pronounced among women. Contrary to concerns over national cohesion, the policy does not exacerbate ethnic tensions and even increases trust in the national institutions. The findings contribute to the literature on large-scale bilingual education and the political economy of language use in Africa, suggesting that integrating local languages into education systems can enhance human capital without undermining nation-building.

JEL Codes: 055, I25, I28

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*"I don't understand" in Bamanakan

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1 Introduction

Learning outcomes in sub-Saharan Africa (SSA) remain alarmingly low despite significant increases in school enrollment over recent decades (UNESCO, 2013). The 2019 PASEC assessment revealed that nearly 60% of students in grade 4 and grade 6 lacked fundamental mathematics and reading skills (CONFEMEN, 2020).¹ This disconnect between school access and actual learning highlights the multiple barriers impeding human capital development in the region. While recent research has primarily addressed supply-side constraints, including inadequate educational resources and teacher capacity (Bold et al., 2017; Glewwe and Muralidharan, 2016), less attention has been devoted to the critical role of the language of instruction in the quality of education.

The use of non-native languages for instruction defines an education system that begins with teachers’ instruction primarily in the vehicular language (i.e. the language used for communication purpose in the community), with a slow transition to the foreign language (often the former colonial language in SSA) until it becomes the exclusive medium of instruction by the end of primary school. A large body of literature in linguistics highlights the benefits of pluri-lingual or bilingual education systems in SSA, reducing the linguistic distance between the school and the home environments for the students (Benson, 2002; Cummins, 2000).² Only a few papers in empirical economics have investigated this issue and causally estimated the effect of using local languages in basic education on learning. For instance, in a small-scale experiment in Cameroon, Laitin, Ramachandran, and Walter (2019) show that grades 1 and 3 students taught in the local language had more than one standard deviation better test scores in mathematics and English. Ramachandran (2017) and Seid (2016) study the introduction of Oromiffa in the primary education system in Ethiopia, instead of Amharic only, and find that mother tongue instruction increases school attendance. Different studies of the PRIMR-MT (Primary Math and Reading in the Mother Tongue) reform in Kenya additionally point that mother tongue literacy increased. However, all these studies provide evidence for only one side of literacy gains: whether learning increased in the familiar language (L1) or in the second language introduced later (L2). Only one recent paper in South Africa provides evidence on the short-term linguistic transfer between the L1 and the L2 (Mohohlwane et al., 2024).

Another tradeoff surrounding bilingual education relates to nation-building. Language unification has been shown to be a key component in nation-building (Blanc and Kubo, 2023; Clots-

¹14 countries were part of the last PASEC (*Programme d’Analyse des Systèmes Educatifs de la CONFEMEN*) evaluation, making it one of the largest international tests in SSA.

²I will use the term bilingual education to refer to both pluri-lingual and bilingual education systems throughout this paper, for clarity purposes.

Figueras and Masella, 2013). Moreover, a study by Pengl, Roessler, and Rueda (2022) documents that ethnic groups in SSA who are more exposed to publication in their language are more correlated with strong ethnic identity feeling. This fear of harming the nation-building process by using ethnic-based languages in education appears to be a strong determinant for policymakers for not introducing bilingual education reform in the SSA education systems.

This paper provides the first empirical evidence on the long-term effects of a national education policy that introduces local languages instead of a colonial language on long-term learning and nation-building. I exploit local variation in the implementation of Mali’s 1999 linguistic education reform, leveraging granular commune-level data to identify the causal effect of bilingual instruction. Mali is one of the last SSA countries where primary school enrollment is still insufficient. Learning levels are also extremely low, with only 30% of the adult population literate in 2020. Until 1999, French was the only language of instruction in primary schools, despite a high linguistic diversity with 63 living languages, including 57 stable and institutionalized (Eberheard, Simons, and Fennig, 2025). In 1999, the Ministry of Education enforced a nationwide reform of the primary education system that introduced 11 national languages as medium of instruction.

Using school-level information on the implementation of the bilingual education (BE) curriculum from the 2011 national census, I estimate the BE supply at the commune level as the share of schools offering BE over the total number of schools. Exploiting the local variation in this ratio at the commune level, I use a difference-in-differences strategy, comparing birth cohorts in high-intensity BE communes to those in low-intensity BE communes before and after the implementation of the reform.³ I rely on the 2018 LSMS survey to have human capital accumulation in different languages, and Afrobarometer data (rounds 8 to 10) for outcomes related to ethnic identity and trust in the institutions.

I find that individuals with greater exposure to the linguistic reform demonstrate a 10% higher likelihood of French literacy and a 30% higher local language literacy in adulthood, highlighting linguistic transfer from L1 to L2. School attendance and success in the primary school leaving exam also increase by 6 and 4 percentage points (pp), respectively. I further show that changes in school supply do not drive these effects, and thus can be attributed to the BE aspects of the reform. I provide suggestive evidence that the positive effects are concentrated on women, as girls are less exposed to the official language before primary education. Moreover, mothers’ education plays a key role in explaining the effectiveness of bilingual education, highlighting an important channel related to parental investment. These findings remain robust across alternative difference-in-differences specifications, including those that relax the assumption of treatment homogeneity or account for variations in treatment timing.

I also look at the political economy of bilingual education, specifically whether introducing local languages in education can have detrimental effects on the nation-building process by promoting ethnic

³I define high and low-intensity BE communes using the median BE provision at the district level, to avoid capturing any district differences.

identities. I show that bilingual education does not harm the creation of a national identity, but rather reinforces trust in institutions, highlighting the absence of a trade-off between improving the quality of education and strengthening the nation-state in a country prone to conflict.

This paper contributes to three strands of literature. First, it provides new empirical evidence on the learning benefits of bilingual education and the linguistic transfers from the familiar to the second language. A recent meta-analysis finds that mother tongue instruction shows promising results in controlled experimental settings (Evans and Mendez Acosta, 2021). This study takes an important first step by examining adult learning and schooling outcomes and suggests avenues for future research on long-term wage effects. While previous studies have mostly focused only on one side of the literacy gains (Laitin, Ramachandran, and Walter, 2019; Piper, Zuilkowski, and Ong’ele, 2016; Ramachandran, 2017; Seid, 2016), this study moves beyond simply looking BE’s impact on a single language by leveraging the richness of LSMS data on literacy across multiple languages.

Second, this paper contributes to the nation-building literature in Africa by examining a policy that promoted the use of local languages. Previous studies on the topics limited their scope to non-African countries (Blanc and Kubo, 2023; Clots-Figueras and Masella, 2013). So far, evidence in sub-Saharan Africa is limited to the study by Pengl, Roessler, and Rueda (2022), which relies solely on correlations at a continental level. I expand on this work by using a strong identification strategy, exposing the causal link between bilingual education and nation-building in a SSA country.

Finally, I contribute to the recent and growing literature evaluating education policies at scale. Implementation is critical to understanding the differential impacts of education programs (Angrist and Meager, 2023). However, scaling up successful interventions remains challenging. Evidence on nationwide bilingual education policies shows mixed results due to implementation difficulties (Piper, Zuilkowski, and Ong’ele, 2016). This study adds to the literature by documenting evidence from a successfully scaled-up policy intervention.

This paper is organized as follows. In Section 2, I document the context surrounding the 1999 reform. Section 3 details the data used for the empirical analysis. I explain the main empirical strategy in Section 4 and present the results in Section 5 for learning and in Section 6 for nation-building.

2 Bilingual education and the 1999 reform in Mali

2.1 Historical background

There is a long-lasting history of bilingual education in the country that started right after the independence.⁴ In 1960, experiments started using local languages to expand literacy for adults who did not attend school under colonial rule (UNESCO, 1963). Following positive results from these experiments, the

⁴During the colonization period, like any other country under the French administration, the official language of education and administration was French, and the use of the local languages in the schooling environment was severely punished (De Gaston, 2011).

use of local languages expanded to primary education at the onset of the 1980s (Ba, 2009; Diarra, 2020).

The first pilots of bilingual education combining French and another local language in primary schools started in the late 1970's, supported by international organizations like UNESCO.⁵ They expanded gradually from Segou to other regions, and from using only Bamanankan to introducing other local languages in education. Although the impact on test scores was limited to null (Maurer, 2007; Skattum, 2010; Traoré, 2001), these experiments provided crucial insights for policymakers during the 1999 expansion of the bilingual education program.

2.2 The 1999 bilingual education reform

Expansion In 1999, the Ministry of Education passed a law to scale up the bilingual education experiment to all public primary schools as an essential component of the PRODEC (*Programme Décennal de Développement de l'Education*), a countrywide educational reform aiming at expanding primary schooling coverage (Loua, 2017). The main objective of this reform was to build schools massively, resulting in a rapid increase in the school supply, from 2,600 schools in 1998 to almost 10,000 in 2008. The bilingual education reform was implemented at the school level at the start of the 2001 school year. In similar contexts, demand has been identified as one of the primary obstacles to the expansion of bilingual education (Piper, Zuilkowski, and Ong'ele, 2016; Ramachandran and Rauh, 2022). However, because of the information campaign that took place in 1999, community demand was high at the onset of the reform.⁶ In 2002, there were 2,110 bilingual classes, and 666 schools with at least one bilingual track throughout the country. The bilingual curriculum was available in 11 additional languages (Bamanankan, Songhay, Tamasheq, Soninke, Dogon, Fulfude, Bomu, Syenara, Mamara, Bozo, and Khassonke) and counted 121,734 enrolled students (MEN, 2003). Following a rapid expansion in the early 2000s, the number of bilingual schools reached a peak of 2,530 in 2005, accounting for approximately one-third of the total number of schools. Since then, a lack of funding and political will has led to stagnation in the number of schools following the bilingual education curriculum. In recent years, this number has even declined, and bilingual education is implemented in only 25% of the primary schools.

Curriculum content Bilingual education is a pedagogical approach that relies on knowledge of the familiar local language to ease the transition to the foreign language (Mohohlwane et al., 2023). In practice, students are expected to spend the first years of primary education learning the basics of writing and reading, as well as other subjects such as Mathematics, in the local vehicular language.⁷ At the same time, teachers introduce French progressively to finish at the end of the primary cycle with French-only

⁵See Section A.1 for more details.

⁶To boost the demand for bilingual education, information sessions were organized at the district level, and short information messages were displayed through 34 local radios (MEN, 2003)

⁷Figure A.1 shows an example of a textbook for grade 1 students, fully written in Bamanankan.

instruction.⁸ During the first two years, the curriculum is primarily in the local language chosen. Over the next two years, French is gradually introduced, so that by the end of grade 4, half of the curriculum is taught in French and the other half in the local language. During the last two years preceding the secondary cycle, French becomes the dominant language. Officially, French remains the sole language of instruction in the second cycle of primary education, as well as in secondary and tertiary education.

Implementation Qualitative evidence points to a demand-led bottom-up implementation process. To open a bilingual education track in a school, local leaders must request approval from the local committee responsible for expanding bilingual education.⁹ In collaboration with the community, this committee would select the primary language to be used in the bilingual class by teachers and students as the language of instruction. Schools would receive textbooks in the selected language, and teachers would undergo a short additional training course to adapt to this new curriculum (20 days in theory) (MEN, 2003).

Official reports document the mixed quality of the policy expansion. In practice, the vast majority of teachers undergo additional training (Diarra, 2013).¹⁰ However, little is known about the training quality, as the bilingual education expansion followed a very decentralized process (Ba, 2009). Reports also point to the long delay in textbook provision, which is attributed to the lengthy printing process (MEN, 2003).

3 Data

3.1 Roll-out of the bilingual education program

To document the progressive expansion of the bilingual education policy throughout the country after its initial implementation in 2001, I exploit a census conducted in 2011 by linguistics experts on behalf of the Ministry of Education (Diarra, 2013). This census covers all schools that were declared officially as “bilingual” at the 2011 school start and reports at the school level whether the bilingual curriculum is still used, or was abandoned for the French-only curriculum.¹¹ Out of the 3,784 bilingual schools that opened since 2001, 83% still had a bilingual education track in 2011.

Due to the civil war at the time of data collection, the census took place only in the Bamako, Kayes, Koulikoro, Segou, and Sikasso regions, covering only one-third of Mali’s territory (see Figure A.4). However,

⁸Table A.1 details the use of French and the local language for each grade, as specified in the official curriculum.

⁹The opening of a bilingual education track in a school can arise from two different situations: either the school is newly built as part of the PRODEC program, or the school switches from the monolingual to the bilingual education curriculum.

¹⁰In 2000, 9 training centers were created. In 2001, 26 centers were operating with 3,775 teachers trained. In 2002, this number increased to 34 centers with 3,608 additional teachers trained (MEN, 2003).

¹¹If the school still has a bilingual class, the census reports whether it is used for all grades or only the first ones. Moreover, information about the local language used at the school level is not available in the census.

as 78% of the population lives in these regions, I argue that this census covers a large fraction of the school supply, reducing the external validity threat for this study (INSTAT-Mali, 2017).

Figure A.2 shows the evolution of the number of bilingual schools from 1994 to 2011.¹² Before the 2000s, the bilingual education supply was close to zero. Following the official introduction of bilingual education into the curriculum in 2000, the proportion of bilingual schools increased progressively to reach 25% of all primary schools.

Qualitative evidence suggests that community schools were more likely than other schools to adopt the bilingual education curriculum (Diarra, 2020). These specific public primary schools are opened as a result of a community initiative; their number rapidly grew during the 1990s as a response to the low school supply. However, using the name of the school in the 2011 census, I find that only 5% of the bilingual schools are community schools.¹³

3.2 Population census

School supply Using census data from 1998 and 2009, I construct a panel dataset on school supply at the commune level.¹⁴¹⁵ Figure A.2 provides an overview of its evolution over time, showing that the number of schools increased rapidly during the 2000s.¹⁶ I use the total school supply information to create an indicator of exposure to bilingual education at the commune level: the ratio between the number of bilingual schools as officially listed in the bilingual education census in 2011 and the total number of schools as provided in the 2009 census.¹⁷

Commune characteristics The main level of the analysis of this study is at the commune level.¹⁸ Using the 2009 census data, I derive some characteristics of communes that I expect to be essential when

¹²Table A.3 details which geographical level, information available, and source used for bilingual education supply for every year presented in Figure A.2.

¹³I consider a school as a community school if it is labeled as such in its name: with the words “EC” or “communautaire”. Hence, this number is a lower bound of the true fraction of community schools among the bilingual education supply.

¹⁴I thank Flore Gubert for providing the panel dataset on the public infrastructures that were used in Chauvet et al. (2015).

¹⁵The administrative system in Mali is decomposed as follows: a region is composed of a set of districts, a district includes different communes, and a commune aggregates various villages.

¹⁶I address this potential confounder and further discuss it in 5.4.

¹⁷If the number of bilingual schools in 2011 is higher than the number of schools provided in 2009, because new schools were created during the 2-year difference for instance, I set this ratio to one.

¹⁸Information scarcity in the data does not allow me to perform the same analysis at the village level. Moreover, even if these data were available, the identification strategy would rely on a strong assumption: that people attend the school in their village and do not attend a neighboring school. Substantial evidence suggests that people are willing to attend other schools to match their language preferences (Ramachandran and Rauh, 2022).

analyzing the efficiency of bilingual education. Specifically, I use the number of speakers for every language at the commune level to derive an ethnolinguistic diversity index. I also exploit the main language spoken in the community as a proxy for the language chosen to be the new language of instruction for the first grades.

3.3 Human capital

Measurement of literacy I am interested in the impact of bilingual education on human capital accumulation. I use literacy as a proxy for education quality. Literacy is a widespread but poorly measured indicator in many surveys, as it is often overestimated. For instance, in the Demographic and Health Surveys (DHS) conducted in developing countries, respondents were automatically considered literate after a certain number of years of education, usually after completing the primary cycle (Sandefur, 2017). However, in many sub-Saharan countries, the Learning-Adjusted Years of Schooling (LAYS) indicates that language proficiency requires more than just finishing primary education when accounting for the quality of schooling (Filmer et al., 2018). Therefore, I use the 2018 Living Standards Measurement Study (LSMS), conducted by the World Bank in 2018, to address this issue. Indeed, the LSMS survey provides a measure of literacy based on writing and reading skills in French, the local vehicular language, and another language for all respondents.

Other outcomes Besides literacy, I also consider other educational outcomes: school attendance, and whether the respondent has the primary school diploma.¹⁹ I do not consider the number of school years as the linguistic literature documents two opposite mechanisms that could affect the time spent at school (Benson, 2002): bilingual education affects the retention rate of students positively, but it is also expected to decrease the repetition rate. Moreover, as the youngest birth cohort in my sample turned 15 in 2018, some may not have completed their education at the time of the survey. Due to data limitations, the completion of primary education is also not considered in this analysis²⁰; instead, obtaining the end-of-primary-school diploma is used as a proxy for completion of the first cycle.

3.4 Nation-building

To investigate the second trade-off highlighted by the literature, I exploit data from the Afrobarometer, using rounds 7 to 10 conducted from 2017 to 2025. This survey is widely used in the political economy literature to study nation-building through ethnic identification (e.g., Pengl, Roessler, and Rueda (2022)).

¹⁹In Mali, primary education starts at the age of 7 officially, for 6 years for the first primary cycle and 3 years for the second primary cycle. At the end of this first cycle, until 2010, students passed an exam called the *Certificat d'études primaires* (CEP) to pass to the second primary cycle.

²⁰In the raw LSMS data, for this survey round in this country only, the variable about education attainment is incomplete.

Ethnic identity Following this literature, I consider outcomes related to ethnic identity, such as whether the respondent feels more Malian than from her own ethnic group. I complement this binary, restrictive outcome by examining other dimensions related to ethnic identity: whether the respondent would like a neighbor from a different ethnicity, as well as whether they feel comfortable wearing traditional clothes or speaking their mother tongue. Indeed, clothing and language are two important determinants of ethnic identification besides self-declared ethnic identity feeling.

Trust in institutions I consider a second set of political economy outcomes, related to trust in institutions. Indeed, the nation-building process involves building reliable institutions that the public can trust. I use trust in the Presidency and the independent electoral commission as proxies for it. Finally, I also treat trust in local institutions as placebo variables, such as traditional leaders and the elected local government council, since any effect on nation-building should not be found at the local level.

4 Empirical strategy

4.1 Identification strategy

Difference-in-difference The provision of bilingual education does not follow a random assignment, and communities that requested the opening of a bilingual education track differ from the others on many observable and unobservable characteristics. Hence, I leverage the introduction of the reform at the 2000 school start and the heterogeneity in the coverage of bilingual education throughout the country to assess its impact on long-term educational and nation-building outcomes. Using the year of birth of adult individuals assessed in the LSMS, I can infer when they started school. In this aspect, if they first attended school before the 2000 school year, it is unlikely that they were exposed to bilingual education (Figure A.2). Given that the average age at school entry in the LSMS sample is 6.6, I consider that individuals born in 1993 and before likely attended monolingual schools (i.e., schools with French-only education). To avoid capturing the effects of other reforms, I restrict the sample to 10 years around the reform implementation, specifically individuals born from 1983 to 2003, aged 15 to 35 in 2018. Then, for each LSMS cluster, I map the supply of bilingual education using the 2011 census data on bilingual schools and 2009 census on school supply, and obtain a panel data set at the enumeration area level with the year of birth as the key time dimension.

Treatment definition I define exposure to bilingual education (BE) at the commune c level as follows:

$$\text{Share of BE}_c = \frac{\text{Nb. bilingual schools}_c}{\text{Nb. schools}_c}$$

I include in the numerator all schools officially declared as bilingual education schools in the 2011 census, and in the denominator the school supply as given by the 2009 census.²¹ By definition, the exposure measure is bounded between 0 and 1.²²

Figure A.5 shows the share of bilingual schools among the total school supply at the commune level, along with the localization of the LSMS cluster. Bilingual education provision is not random and follows a clear geographical pattern, with some districts and regions concentrating most of the bilingual education supply. For instance, looking at the linguistic areas depicted in Figure B.7, I observe that the supply of bilingual education is high in locations where Bamanankan is the main language. As a result, I cannot use the national median exposure to bilingual education, i.e., the fraction of bilingual schools in the total school supply, as a threshold to categorize communes into low vs. high intensity, mimicking the design used in Duflo (2001). To overcome this issue, I use the median ratio of bilingual schools over the total number of schools within each district as a threshold and consider a commune with a ratio below this median to be a low-exposure area.²³ Figure 1 displays the geographical coverage of the binary treatment.

Sample selection One concern about the sample considered is that migration already took place at the time of the survey. In 2011, migration, both internal and external, was estimated to affect almost 20% of the population (Sougane, 2014). It may impact my empirical specification in two ways. First, I do not observe in the LSMS sample migrants who left the locality to live abroad. However, if anything, this should yield a downward bias in the results: if bilingual education enhances learning skills, particularly in French, then individuals who benefited significantly from the bilingual curriculum leveraged these new skills to migrate before the LSMS survey took place. Hence, the estimated effect would be a lower bound of the true impact of bilingual education. Second, I may wrongly attribute a high exposure to bilingual education to an individual who migrated after their childhood. The level at which the analysis is done reduced this threat: 98.8% of individuals in the sample are reported being born in the same commune.

4.2 Empirical specification

Two Way Fixed-Effects (TWFE) By leveraging the reform implementation timing and the differential exposure to bilingual education, I estimate the following regression:

$$Y_{i,y,c,d} = \alpha_1 \mathbb{1}[y \geq 1994]_{i,c,d} + \alpha_2 \mathbb{1}[c \in \text{BS}]_{i,c} + \beta \mathbb{1}[y \geq 1994]_{i,c,d} \times \mathbb{1}[c \in \text{BS}]_{i,c} + \theta_d + u_{i,y,c,d} \quad (1)$$

$Y_{i,y,c,d}$ is the outcome for individual i , born in year y , living in the commune c located in the district

²¹I further consider the fraction of schools that dropped out of the bilingual curriculum in the section 5.4.

²²Because the school supply is measured through the population census that took place 2 years prior to the 2011 bilingual education census, some ratios exceeded 1 for a few communes. For these communes, I bounded the ratio to 1.

²³Similarly, I consider a commune with a fraction of bilingual education higher than the district median as highly exposed to bilingual education.

d. *BS* represents the set of communes considered as highly exposed to bilingual education relative to the median district ratio of bilingual schools. α_1 captures the effects of being born after 1994, i.e., starting school after the implementation of the bilingual education reform. α_2 isolates the differences in the outcome that might exist between communes with a high share of bilingual schools compared to communes with a low share. The coefficient of interest is β , which captures the effect of being highly exposed to bilingual education (BE) compared to low exposure after the reform. I use district-fixed effects θ_d to control for district unobservable characteristics and differences in bilingual education supply (Figure A.5). I use the LSMS cluster provided in the survey to cluster standard errors, and I use household weights also present in the survey.²⁴

Testing key assumptions Following the difference-in-difference literature, I test for the parallel trends (PT) and the no anticipation (NA) assumptions using an event study design. To do so, I estimate the following regression:

$$Y_{i,y,c,d} = \sum_{t=1983}^{2003} \alpha_{1,t} \mathbb{1}[t = y_{i,c,d}] + \alpha_2 \mathbb{1}[c \in \text{BS}]_{i,c} + \sum_{t=1983}^{2003} \beta_t \mathbb{1}[t = y_{i,c,d}] \times \mathbb{1}[c \in \text{BS}]_{i,c} + \theta_d + u_{i,y,c,d} \quad (2)$$

The same definitions as in Equation (1) apply. t represents the year of birth and covers the period considered in this analysis, i.e. from 1983 to 2003. I allow the coefficient β_t to vary in the pre and post-treatment period according to the year of birth y . I also implement multiple robustness tests in Section 5.4 to reinforce the credibility aspect of the PT and NA assumptions.

An additional key assumption made in the specification is the homogeneity of the treatment effect. This assumption is likely not to hold, as the treatment at the commune level is relative to a median value at the district level.²⁵ I employ different strategies to show that this is unlikely to bias my estimate. First, I control for this by adding a fixed effect at the district level in Equation 1. Second, I release the assumption of treatment effect homogeneity in Section 5.4 by employing different approaches.

4.3 Descriptive statistics

Table 1 describes the main outcomes for the sample, at the individual and the commune level (Columns 1 to 3). The sample includes 8,636 individuals, half of whom are women and half of whom are urban. Less than 50% of the sample attended school, and the same fraction is literate in French (either writing

²⁴Given that a commune is roughly equal to one LSMS cluster (except for Bamako), and that Bamako represents 60 clusters out of 325, I prefer using the LSMS cluster level rather than the commune level for clustering.

²⁵In other words, I assign commune A in district 1, where 70% of schools are bilingual, to the low-exposure bilingual education group. In contrast, commune B in district 2, where only 20% of schools officially use local languages for instruction, can be assigned to the high-exposure bilingual education group. This apparent contradiction arises because the median ratio of bilingual education supply is 75% in District 1 but only 15% in District 2.

or reading). Finally, only 1 out of 10 individuals can write or read in the local language. Columns 4 to 6 provide additional insights into the significant differences between the sample of communes less exposed to bilingual education and those highly exposed to it before the reform implementation. The main difference arises from local language proficiency: individuals in locations with more bilingual schools after 2001 are 3pp more literate in the local language. This “balancing” further highlights the non-random assignment of bilingual education throughout the country and the need for a quasi-experimental design.

5 Results on learning

In this section, I investigate the effects of the bilingual education reforms on learning. Then, I estimate the differential effects according to students and parents’ gender before looking at additional set of mechanisms.

5.1 Literacy in French and in the local language

Figure 2 plots estimated coefficients from Equation 2 explaining BE effects on learning outcomes in the local language. The first result is that the low- and high-BE exposure groups follow parallel trends before the implementation of the 1999 bilingual education reform. Moreover, I observe that introducing officially local languages in education has a significant, positive, and persistent impact on writing and reading in the vehicular language. Table 2 shows the TWFE results of estimating Equation 1: adults who had access to more bilingual schools when they were children are 4pp more likely to be literate (both writing and reading) in their local language (Columns 1 and 2). Given that only 13% of the sample is literate in the local language, this translates into approximately 30% learning progress. These initial sets of results can be viewed as a first stage: even though it is likely that teachers had already used local languages before their introduction into the curricula in 1999, their official enactment increased *de facto* their use in the schooling environment, thereby boosting literacy in the local language.

Figure 3 shows that these skills in the local languages partially transferred to French for the exposed students, in line with linguistics literature Cummins (2000). Columns (3) and (4) of Table 2 provide ATE estimates: on average, French learning increased by 5pp (i.e., a 10% increase of the literacy rate), both in writing and reading, with the introduction of the bilingual education reform. These results are less significant and smaller in magnitude than those for the local language, as some skills may not have been fully transferred from one language to the other because the children attended school for only 4 years. These results confirm the short-term evidence provided by Mohohlwane et al. (2023) in the South African context: increasing learning in the local language does not come at the cost of learning in the second language in a bilingual education system, but rather the opposite.

In addition to literacy in the two languages, LSMS data provide the most appropriate placebo variable: literacy skills in a language neither French nor the community’s main language. Introducing bilingual education should not affect this outcome at all, as this language is unlikely to be used in bilingual schools

in this area. Figure B.2 shows the event-study estimate and provides additional evidence supporting the identification strategy, as I do not observe any change in writing and reading skills after introducing bilingual education.

Introducing familiar languages to students makes school more attractive: as shown in Column (1) of Table 3, school attendance increased by 12% for cohorts that got access to more bilingual schools (+ 6pp). They are also more likely to hold a primary education diploma, suggesting that the duration of schooling is also affected. Same results are documented in the linguistics literature, which shows that reducing the linguistic distance between the home and school environments incentivizes more parents to send their children to school (Benson, 2004).²⁶

5.2 Gender

A linguistic study by Benson (2005) documents that bilingual education helps girls catch up to boys. Figure 4 shows heterogeneity results by gender and provides evidence supporting this qualitative result.²⁷ Women who had more access to bilingual schools as children are 8pp more likely to be literate in French and attend school.²⁸ They are also 6pp more likely to obtain the end-of-primary-school diploma. Relative to the sample mean, it represents respectively an increase of 15% in the probability of being literate in French and attending school, and 20% in obtaining the CEP. In comparison, boys do not get any returns from introducing education in a familiar language. Figure B.4 displays graphically how girls partially catch up to boys thanks to bilingual education.

I explore two mechanisms explaining this result. The first one relates to the exposure to French prior to entering school, qualitatively documented in the study by Benson (2005). Using the UNICEF Multiple Indicator Cluster Surveys (MICS) conducted in Mali in 2015, I show in Table B.1 that girls are already less familiar with the French alphabet in their first years (Column 1) and work less outside of the household before starting school (Column 3). This differential exposure to French partly explains why using local languages benefits the girl students more.

I explore a second mechanism at the maternal investment level, related to a recent paper by Dizon-Ross and Jayachandran (2023).²⁹ They show, through a field experiment in Uganda, that fathers' willingness-to-pay (WTP) for their daughters' education is much lower than for their sons', whereas mothers' WTP does not differ. Thanks to the introduction of local languages at the school level, mothers are expected to

²⁶Parallel trends for the last set of outcomes are shown in Figure B.1 and confirm that both groups follow the same trends before the implementation of the bilingual education law.

²⁷Parallel trends at the subgroups levels are shown for literacy outcomes in Figure B.3. No anticipation effect is observed for either women or men. Event studies for other education outcomes give similar conclusions.

²⁸The differential bilingual education effects for women and men are significantly different at the 5% level.

²⁹Heterogeneity analysis at the parental level already highlights that mother's education is a key component for bilingual education efficacy (Figure B.5).

be more engaged in their children’s education. As their level of engagement increases, the investment bias (money, time, etc.) toward sons is reduced, benefiting girls. Using 2018 DHS data, I provide suggestive evidence that mothers in communes highly exposed to bilingual education gain some decision-making power (Table B.2). Specifically, the likelihood that the husband has the final say on visiting people outside the household decreased by 12pp, whereas coefficients related to other decisions are negative, although imprecisely estimated. These two mechanisms shed light on language exposure and maternal investment as two potential reasons explaining why bilingual education can close the gender gap, but more research is needed on this specific point.

5.3 Local conditions

Linguistic diversity I expect the impact of bilingual education to be proportional with the share of students whose mother tongue aligns with the chosen new language of instruction. To test for it, I stratify the sample by quartiles of linguistic diversity at the commune level, measured using a Herfindahl-Hirschman index derived from the 2009 census data on language use.³⁰ Contrary to expectations, empirical results in Figure B.6 indicate that high linguistic diversity does not impede the effectiveness of bilingual education, highlighting interesting findings for policy implications.

Dominant language The selection of the instructional language may still play a critical role, independent of overall linguistic diversity. As illustrated in Figure B.7, Bamanankan, spoken as a first language by 4 million Malians and as a second language by an additional 10 million, serves as a *lingua franca* across linguistic communities. However, regression results in Table B.3 (Panel 1) reveal that the positive effects of bilingual education are not driven by Bamanankan-dominant areas. Thus, neither linguistic diversity nor the dominance of a particular language emerges as a decisive factor in the success of bilingual education programs.

Urbanization I further explore whether rural households derive greater benefits from local-language instruction. Indeed, urban areas tend to use French more as a vehicular language between the different ethnolinguistic groups gathered in the same place. Panel (2) in Table B.3 presents mixed findings: local language proficiency improves only in urban areas, with no discernible effect on French literacy. To address potential endogeneity in urban classification, since urban status in the 2018 LSMS survey may reflect changes induced by the reform, I refine the analysis by using historical road access as a proxy for urbanization. Drawing on road network data from Müller-Crepon (2023) and Müller-Crepon, Hunziker, and Cederman (2021), I measure each cluster’s distance to the nearest road as of 2000, applying a 10-kilometer threshold (see Figure B.8).³¹ Heterogeneity analysis, reported in Table B.3 (Panel 3), shows that the

³⁰Specifically, linguistic diversity is calculated as $1 - \sum s^2$, where s represents the share of speakers for each language reported in the census.

³¹Urban status and proximity to roads in 2000 are positively and significantly correlated (coefficient: 0.47, SE: 0.05). Importantly, the treatment variable for bilingual education is uncorrelated with baseline

positive impacts of local-language instruction on both learning and schooling outcomes are not concentrated in areas further from roads. Moreover, local language proficiency seems to improve in locations with better road access. These results highlight potential unequal returns to bilingual education, particularly for rural households.

5.4 Robustness checks

Continuity in the bilingual education curriculum Table B.3 (Panel 4) presents a comparative analysis of the effects of bilingual education between communes where at least one school abandoned the bilingual program and those where schools maintained the implementation of the language reform.³² Of the 167 communes analyzed, 100 experienced at least partial reversion from bilingual to monolingual (French-only) instruction in their schools. As expected, positive learning and schooling outcomes are concentrated in areas with a continuous implementation of the linguistic education reform.

Relaxing the homogeneous treatment effect assumption Recent advances in the difference-in-differences literature highlight that the validity of TWFE estimators relies on treatment effect homogeneity (Callaway and Sant’Anna, 2021; de Chaisemartin and D’Haultfœuille, 2022). This assumption is unlikely to hold in the present context. Although newer approaches allow for continuous treatments (Callaway, Goodman-Bacon, and Sant’Anna, 2024), they are not well-suited here because I do not rely on panel data at the individual level to estimate the effects. Moreover, in this setting, comparing treatment intensity would mostly capture geographic differences rather than causal effects, as shown in Figure C.5.

To relax this assumption, I first examine heterogeneity by the district-level median share of bilingual schools used to define treatment. Splitting the sample into quartiles based on this district’s median, I find that treatment effects are generally null or positive across all groups, but vanish in the top quartile, where bilingual education is nearly universal (Figure C.1). In such areas, the lack of meaningful variation limits the comparison between treated and control communes.

I further exploit variation in distance from the district median used to discretize the bilingual education treatment. Communes far from the median cutoff used to define treatment exhibit stronger effects, whereas those close to it show smaller and insignificant estimates, as shown in Table C.2. This pattern suggests that treatment and control groups are less differentiated near the threshold, weakening identification. All these findings underline that relaxing the homogeneity assumption yields similar conclusions.

School construction The main objective of the PRODEC reform was to increase school supply by massively building school infrastructures. Figure C.6 maps the growth rate of the school supply at the commune level: the building effort was widespread across the Southern regions of Mali, and did not target one specific area. Expansion of school supply can be a confounder of the impact of bilingual education, road access.

³²The median switching rate at the potential cutoff point is zero, making the intensive and extensive margin analyses equivalent in this context.

as the 1999 reform included both components. I test this hypothesis by first looking at the correlation between bilingual education supply and the growth rate of school supply in Figure C.2, using the number of schools in 1998 and 2009 as reported in the census. I do not see any evidence that communes with more schools built during this period also had more bilingual schools. Second, I estimate Equation 1 with the number of schools as the outcome (Table C.4). I find that in communes with more bilingual schools, fewer schools were built from 1998 to 2009. Hence, I interpret the previous findings as a lower bound on the true impact of bilingual education, given that these areas were relatively less well equipped with school infrastructure over time. This evidence suggests that the two components of Mali’s 1999 education reform, bilingual curriculum and school construction, were implemented in different areas, rather than as complementary across the country.

Finally, I look at the differential effects of bilingual education intensity across communes with varying exposure to school construction programs. I find that introducing a bilingual curriculum was effective where fewer schools were built during the first decade of the 2000s (Table C.3, Panel 1). This evidence suggests that the benefits of bilingual education do not stem from increased school infrastructure, but rather from the program itself.

Alternative treatment definitions The baseline treatment is defined relative to the district median to account for the unequal distribution of bilingual education. As an alternative, I use an absolute cutoff: whether the share of bilingual schools exceeds 50%. Because most districts in Western and Eastern Southern Mali fall below this threshold, I restrict the sample to central districts, where Bamanankan is the dominant language (Figure C.3). Estimating Equation 1 with this definition (Table C.6) yields similar results overall, except for local language literacy, where effects are null and precisely estimated. This likely reflects the widespread prior knowledge of Bamanankan as a *lingua franca*.

I then exploit district-level variation in the expansion of bilingual education using a staggered DiD with the imputation estimator of Borusyak, Jaravel, and Spiess (2024).³³ Treatment timing is defined as the year a district exceeds its historical median share of bilingual education provision (1994–2005). Although estimates are less precise due to the smaller sample (28 districts), results confirm parallel pre-trends and show positive effects on writing skills in both French and local languages (significant at 10%), with larger impacts for the latter.

Finally, I redefine treatment using the regional (rather than district) median and include region fixed effects (Table C.1). Results remain largely unchanged, though standard errors increase. The only notable difference concerns local language outcomes, again suggesting that treated individuals may already have been proficient prior to the reform.

³³Following the procedure suggested in Roth et al. (2023), I prefer the imputation estimator over the Callaway and Sant’Anna (2021) as event studies displayed in Figures 3, 2 and B.1 show that we should not worry about violation of the parallel trends assumption, and I do not expect important serial correlation between birth cohorts.

Other robustness tests Table C.5 assesses robustness to additional controls, including the ratio of school-age children (ages 7–12)³⁴ to the number of schools (Panel 1) and the number of schools in the commune (Panel 2), as well as linguistic-area fixed effects (Panel 3). Results remain stable across specifications. Going further, to address potential spillovers across communes, I exclude areas within 2 km of borders (Table C.7). Although estimates lose statistical significance due to reduced power, coefficients remain very similar, supporting the initial findings. Next, I explore heterogeneity by migration status (Table C.3, Panel 2). Positive effects are concentrated among non-migrants, while migrants exhibit no significant gains, suggesting that continuous exposure to bilingual education is key to its effectiveness. Additionally, heterogeneity analysis by attendance status (Table C.3, Panel 3) confirms that the estimated effects are driven by individuals who attended school. However, since school attendance is endogenous to treatment, these results should be interpreted cautiously. Finally, I test for potential confounders by examining whether treatment affects migration or school entry age. Table C.8 shows no evidence of such relationships, alleviating concerns about selective migration or differential timing of school enrollment.

6 Results on nation building

This section provides novel evidence on the impact of bilingual education on nation-building in a Sub-Saharan African context, and examines whether the use of local languages in education may hinder the formation of a shared national identity.

6.1 Ethnic identity

To assess the salience of ethnic identity, I estimate Equation 1 augmented with Afrobarometer survey round fixed effects. Rather than focusing exclusively on the binary opposition between ethnic and national identification, I consider a broader set of outcomes capturing the multidimensional nature of ethnic identity. These include both exclusionary attitudes, such as whether respondents report disliking neighbors from other ethnic groups, and markers of cultural expression, such as comfort with wearing traditional clothing or speaking one’s mother tongue in public.³⁵

Table 4 presents the results. Overall, the introduction of local languages into the Malian education system does not lead to an increase in politically salient ethnic identity. If anything, the estimates in Columns (1) and (2), although imprecise, suggest a decline in exclusionary ethnic attitudes. Importantly, this pattern does not reflect a suppression of cultural identity. On the contrary, results in Column (3) indicate that individuals more exposed to bilingual education during childhood report 8pp more comfort expressing their cultural identity in public.

³⁴The number of children theoretically at school if school attendance were complete is computed as the number of individuals aged between 7 and 12 in the 2009 census.

³⁵For interpretation purposes, all outcomes are dichotomized. Details are provided in the notes of Figure 5 and Table 4.

Taken together, these findings suggest that bilingual education may decouple cultural expression from political ethnic identification: individuals retain (or even strengthen) cultural attachment to their ethnic group without translating this attachment into exclusionary or politically salient ethnic preferences.

6.2 Trust in institutions

Next, I examine whether the expansion of bilingual education in multiple local languages affects trust in public institutions. Figure 5 distinguishes between national institutions (the Presidency and the national electoral commission) and local institutions (elected local government councils and traditional leaders). The shaded areas represent estimated effects across outcomes.

Exposure to bilingual education is associated with a statistically significant increase in trust in national-level institutions, with effect sizes ranging from 7 to 9pp. In contrast, while coefficients for trust in local institutions are positive, they are not statistically distinguishable from zero. These results complement the findings on ethnic identity. Bilingual education appears to strengthen nation-building by increasing confidence in national institutions, without crowding out trust in local or traditional structures. These results draw a global picture in which national and local identities are not substitutes but can coexist.

A plausible mechanism underlying these effects relates to improvements in learning outcomes. Civic education is formally included in the Malian curriculum from primary through secondary school, covering basic knowledge about the state and its institutions. In addition, subjects such as history and geography contribute to the construction of a shared national narrative. By improving comprehension through instruction in local languages, bilingual education likely enhances students' ability to absorb this content. As a result, individuals develop a better understanding of national institutions and exhibit greater trust in them, while maintaining their ethnic and cultural identities.

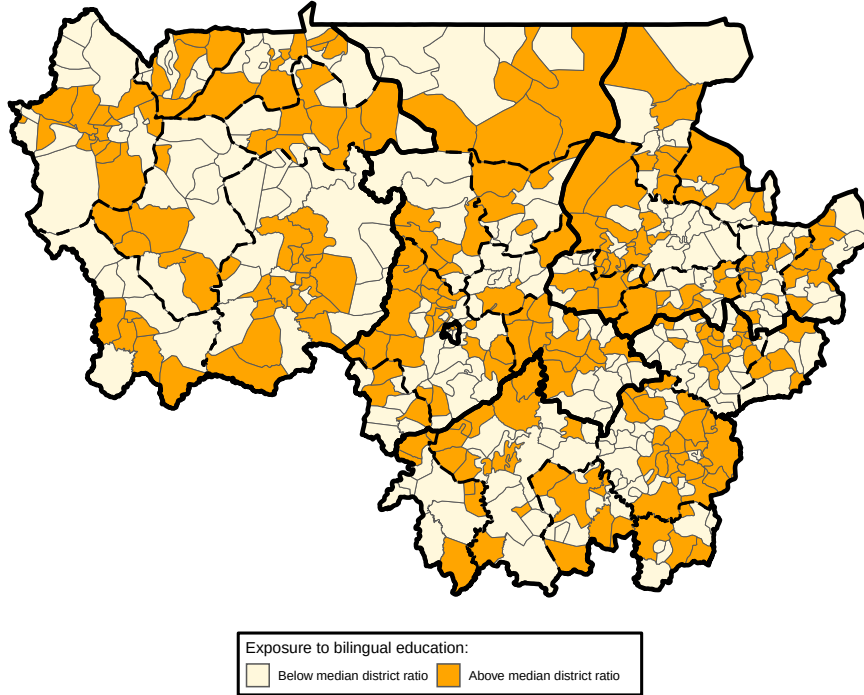
However, the analysis is subject to data limitations. The Afrobarometer data lack detailed information on individual educational trajectories (e.g., school enrollment), which constrains the scope for heterogeneity analysis and robustness checks, in contrast to the analysis in Section 5. Future research leveraging richer educational data would help further unpack the mechanisms linking local language instruction, identity formation, and institutional trust.

7 Conclusion

This paper estimates the impacts of bilingual education in Mali on learning in different languages and nation-building using implementation data on the current bilingual education program. The impacts are high on learning in the local language and in French. Girls are the ones who benefit the most from these additional new languages of instruction. All of these learning gains do not come at a cost to nation-building, and I even highlight a positive effect on trust in national institutions. Section D provides insights into the monetary cost of such an intervention. While very preliminary, these computations suggest that bilingual education is very cost-effective.

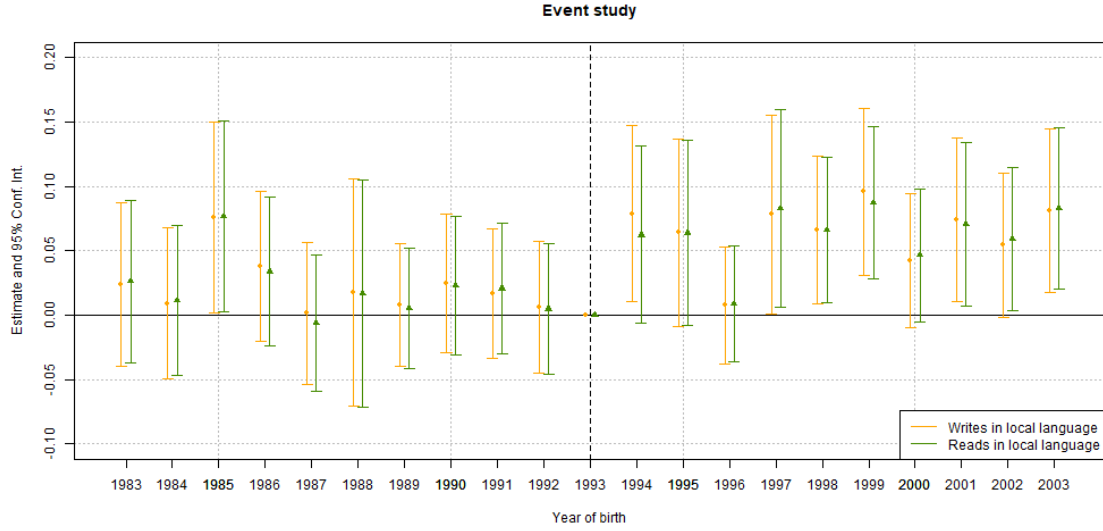
These findings have strong policy implications. In particular, results in Section [5.3](#) suggest that not all individuals benefit equally from the use of local languages. Further research is needed to determine whether bilingual education includes the most marginalized populations, which would provide policymakers with more precise estimates of its benefits across various implementation contexts.

Main Figures



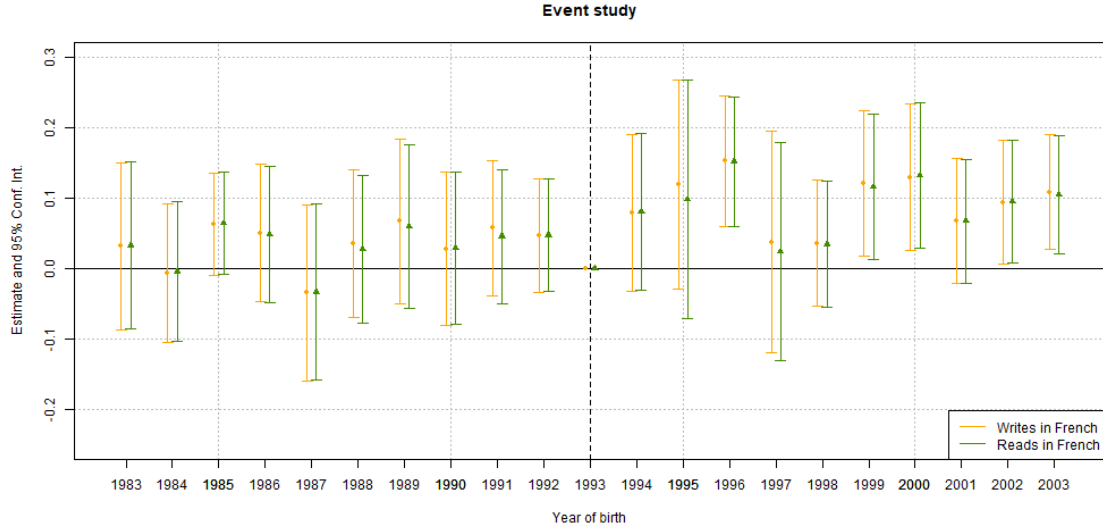
Notes: Solid black lines represent the regional borders, dashed black lines represent the district borders, and gray lines represent the communal borders. The two colors indicate the treatment status at the commune level, as defined in Subsection 4.1. This figure maps only the following regions that are considered in the empirical analysis: Bamako, Kayes, Koulikoro, Segou, and Sikasso.

Figure 1: Map of the treatment status by commune



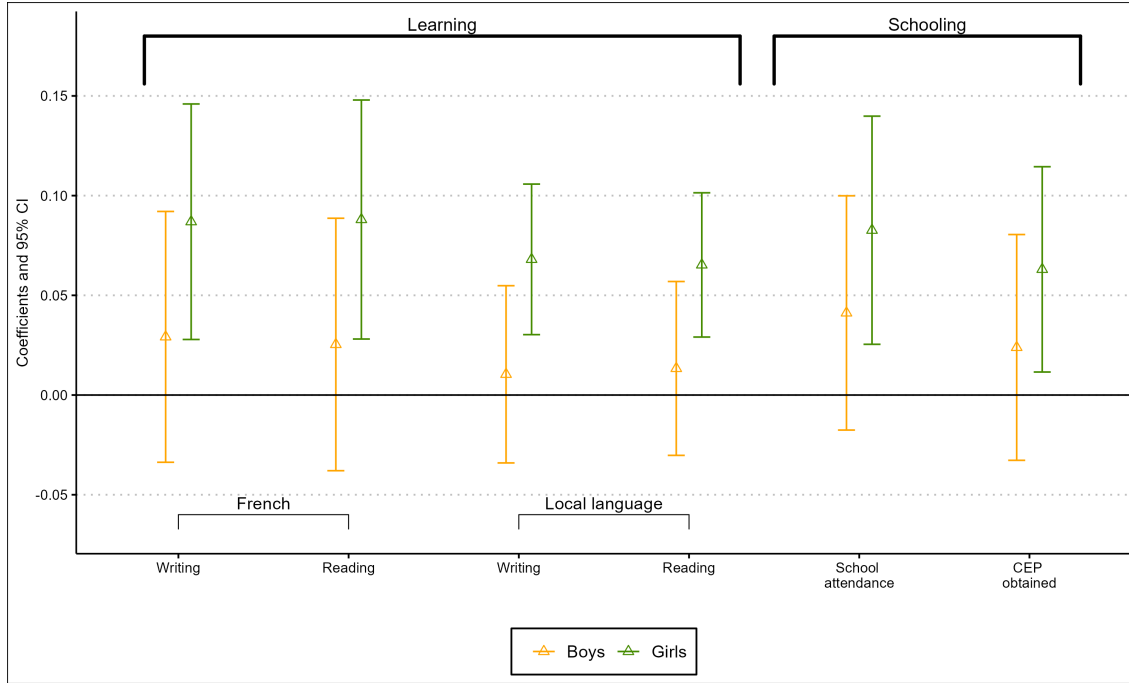
Notes: Point estimates and 95% confidence intervals are derived from an event study regression over the 1983-2003 birth cohorts from which I extracted only the estimates of the interaction terms. I use district fixed effects to capture variation in the median of bilingual education share across districts. To obtain these estimates, I perform two distinct regressions: one with whether the individual knows how to write in the local language as the dependent variable (in orange), and one with whether the individual knows how to read in the local language as the dependent variable (in green). I cluster the standard errors at the LSMS cluster level, which corresponds roughly to the commune level. Individuals born in 1993 are the last birth cohort to be taught entirely in French (i.e., not treated) represented by a black vertical dashed line.

Figure 2: Event study regression for literacy in the local language



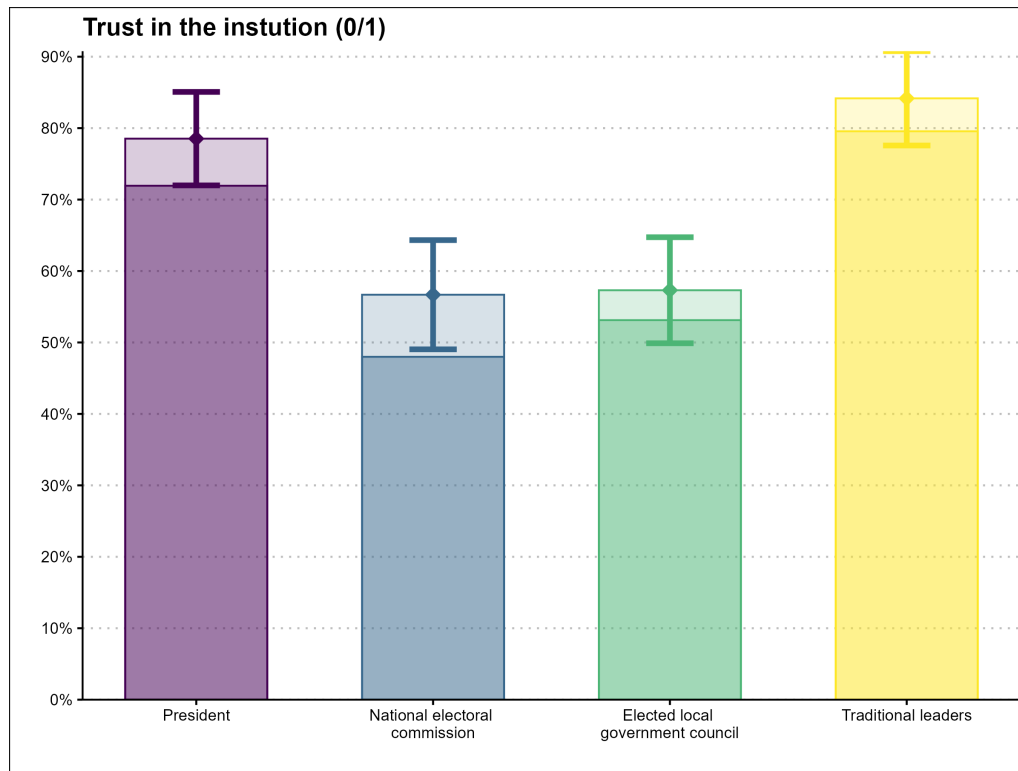
Notes: Point estimates and 95% confidence intervals are derived from an event study regression over the 1983-2003 birth cohorts, specified in Equation 2. In addition to time fixed effects, I use district fixed effects to capture that the median of bilingual education share varies across districts. To obtain these estimates, I perform two distinct regressions on two separate outcomes: (i) whether the individual knows how to write in French as the dependent variable and (ii) whether the individual knows how to read in French as the dependent variable. I cluster at the LSMS cluster level, which corresponds roughly to the village level. Individuals born in 1993 are the last birth cohort to be taught entirely in French, i.e., not treated, represented by a black dashed vertical line.

Figure 3: Event study results for literacy in French



Notes: Point estimates and 95% confidence intervals are derived from a TWFE linear regression on several outcomes: (1) and (2) are writing and reading literacy in French, (3) and (4) in the local language, and (5) (6) and (7) are school achievements indicators. I use district fixed effects to capture the fact that the median of bilingual education share varies across districts. I cluster at the LSMS cluster level, which corresponds roughly to the village level. To obtain heterogeneous effects, I interact the DiD coefficient in Equation 1 with sex.

Figure 4: TWFE coefficient estimates of bilingual education effects, by sex



Notes: The solid bars represent the outcome mean for birth cohorts that went to school before the implementation of the reform, for the four outcomes described on the x-axis. The shaded areas above the solid bars illustrate the effect of bilingual education on each of these outcomes, with the corresponding 95% confidence intervals. Point estimates and 95% confidence intervals are derived from a TWFE linear regression, described in Table 4 to which I add Afrobarometer round fixed effects. The outcomes are derived from questions Q41A (President), Q41C (National Electoral Commission), Q41D (elected local government council) and Q41K (traditional leaders) in Afrobarometer round 8 codebook. All outcomes ask about the level of trust that the respondent puts in an institution. I coded "somewhat" and "a lot" as 1, and "not at all" and "just a little" as 0.

Figure 5: TWFE coefficient estimates of bilingual education effects on trust in the institutions

Main Tables

Characteristics	Mean	SD	N_1	Diff.	p-value	N_2
in means						
A. Individual characteristics						
Female (0/1)	0.55	0.5	8636	-0.04	0.03	4200
Age	23.7	6.12	8636	0.13	0.27	4200
Muslim (0/1)	0.95	0.22	8636	-0.02	0.2	4200
Urban (0/1)	0.49	0.48	8636	0.06	0.3	4200
Attended school (0/1)	0.51	0.5	8635	0.04	0.25	4199
Number of schooling years	4.01	6.25	8636	0.56	0.27	4200
Literate in French (0/1)	0.52	0.5	8633	0.06	0.17	4198
Literate in the local language (0/1)	0.13	0.33	8623	0.03	0.03	4195
B. Commune characteristics						
Number of students in school age	616.65	893.63	167	-32	0.74	167
Number of primary schools	24.74	35.01	167	-8.54	0.04	167
Share of bilingual schools	0.45	0.32	167	0.28	0	167
Linguistic HHI	0.46	0.15	165	0.01	0.76	165

Notes: (0/1) indicates a dummy variable. “SD” stands for standard deviation. All descriptive statistics are computed using weights provided in the LSMS survey. N_1 stands for the number of non-missing observations when computing these descriptive statistics. Column (3) presents the difference in means between the high-exposed and low-exposed to bilingual education sample (individuals and communes). Column (4) displays the p-value associated with the mean difference. The coefficients and p-values are estimated using weights provided in the LSMS survey, on the subsamples of individuals born before 1994, and clustering standard errors at the survey location level for individuals. The size of the sample used for estimating the mean difference between high and low-exposed samples is described in Column 5, N_2 . Fixed effects at the district level are used in all these estimations.

Table 1: Description of the sample

Language tested	Local language		French	
	Writing	Reading	Writing	Reading
Dependent Variables:				
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Born after 1994	0.008 (0.012)	0.007 (0.011)	0.138*** (0.020)	0.137*** (0.020)
High exposed commune	0.023 (0.015)	0.022 (0.015)	0.034 (0.037)	0.032 (0.037)
Born after 1994 $\times$ High exposed commune	0.039** (0.018)	0.039** (0.017)	0.055* (0.028)	0.054* (0.028)
<i>Fixed-effects</i>				
District FE	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Mean of Y	0.123	0.118	0.489	0.484
Observations	8,636	8,623	8,636	8,633
R ²	0.04228	0.04123	0.15238	0.14913

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw LSMS data. I use the weights given in the LSMS data, and cluster the standard errors at the enumeration area. Clustering at the treatment level, i.e. the commune, produces the same results.

Table 2: TWFE results on learning outcomes

Dependent Variables:	Attended school	Obtained primary school diploma
Model:	(1)	(2)
<i>Variables</i>		
Born after 1994	0.156*** (0.019)	0.073*** (0.016)
High exposed commune	0.023 (0.032)	0.028 (0.032)
Born after 1994 $\times$ High exposed commune	0.059** (0.026)	0.041* (0.025)
<i>Fixed-effects</i>		
District FE	Yes	Yes
<i>Fit statistics</i>		
Mean of Y	0.495	0.305
Observations	8,635	8,635
R ²	0.12179	0.12921

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw LSMS data. I use the weights given in the LSMS data, and cluster the standard errors at the enumeration area. Clustering at the treatment level, i.e. the commune, produces the same results.

Table 3: TWFE results on schooling outcomes

Dependent Variables:	Doesn't dislike having a neighbor from another ethnic group	Feels more Malian than from her ethnic group	Feels comfortable... Wearing her traditional or cultural dress in public	Speaking her mother tongue in public
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Born after 1994	-0.033* (0.018)	-0.002 (0.021)	-0.036 (0.026)	-0.044 (0.032)
High exposed commune	0.013 (0.016)	-0.024 (0.026)	-0.051* (0.028)	-0.037 (0.031)
Born after 1994 $\times$ High exposed commune	0.029 (0.024)	0.035 (0.033)	0.077** (0.038)	0.077 (0.049)
<i>Fixed-effects</i>				
District FE	Yes	Yes	Yes	Yes
Survey round FE	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Mean of Y	0.932	0.859	0.955	0.943
Observations	1,937	1,930	466	467
R ²	0.03582	0.03457	0.10228	0.07983

Clustered (Afrobarometer EA) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The first outcome is derived from the question Q86B, and the second from the question Q82B, as coded in the codebook for the round 8 of Afrobarometer; I assigned a value of 0 for answers between 1 and 2, and 1 for answers between 3 and 5. The third outcome is derived from the question Q82D and the fourth from the question Q82C. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw Afrobarometer data. I use the weights given in the Afrobarometer data, and cluster the standard errors at the enumeration area for every round of survey data collection. Clustering at the treatment level, i.e. the commune, produces the same results.

Table 4: TWFE results on ethnic salience outcomes

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Appendix

A Context

A.1 History of bilingual education prior to the 1999 policy

First-generation experimental schools In 1979, four schools opened a class using bilingual education in Bamanankan only, the main vehicular language in Mali (Calvet, 1993; Traoré, 2001). Teachers used the local language only in the first four years of primary education. Positive evaluations of the bilingual program at the beginning of the experiment led to its rapid expansion. A few years later, almost 100 schools operated in the four main languages: Bamanankan, Songhay, Tamasheq, and Fulfulde (Hutchison, Diarra, and Poth, 1990; Skattum, 2010). Lack of monitoring, teacher training, and sufficient budget led to the end of the first experimentation at the beginning of the 1990s (Skattum, 2010).

The *Pédagogie Convergente* This new experiment, developed and piloted by a Belgian linguistics center, the CIAVER (Centre International Audiovisuel d’Études et de Recherches), started in 1987 with two classes in Segou in Bamanankan. Following a positive evaluation of the pilot, it rapidly expanded in 1994 to other languages and had grown to approximately 100 schools by 1997 (Traoré, 2001). Unlike the first experimental schools, the 6 years of primary education were entirely taught in the local language, with a gradual introduction to French once students had fully mastered their mother tongue.

Qualitative and quantitative evaluations conducted during the experiment provided mixed evidence about its results (Skattum, 2010; Traoré, 2001). Maurer (2007) also reported implementation issues that hindered the program’s scale-up.

A.2 Figures

Bolofara !

Kumajaganya

Sugulataa

1 Buba : Ali, tile koratara de ! O tuma i jinena wa ?

2 Ali : Ala ko tije. fo n'i ye n hakili jigin.

3 Buba : Ne tun m'a fo n be taa sanbara san sugu la bi ?

4 Ali : Ee ! Hakili ye jon ye. N'i labennen don, an ka taa.

5 Buba : Sugu in sanbarako ka gelen waati daw la de !

6 Sanba : N'teri, o ye tije ye. Geleya be fan bee fe sisan.

7 Sanba : Ali ! Buba ! A' dun senna ka teli !

8 Ali : An be taa sanbara jini Buba ye.

9 Sanba : Baara b'a' bolo ! Ni wari caman t'a' kun, hali a' ka segin so.

10 Buba : Ko ten wa ? Jaa an na segin korolen ma !

Bolofara !

11 Ali : Buba, i ma Sanba ka kuma men ?

12 Buba : A ka nogo Ala fe. N kunta mana min soro, n n'o san ten.

13 Sanba : A' k'an ben ! N be taa n ba nofe dagafeeredankan na.

14 Ali : Ne yere be taa ji di baw ma.

15 Buba : An be pagon soro ntolatanyoro la wula fe.

16 Sanba : Bi ta koni te fo n ko. N be se yen bee ge.

Daga

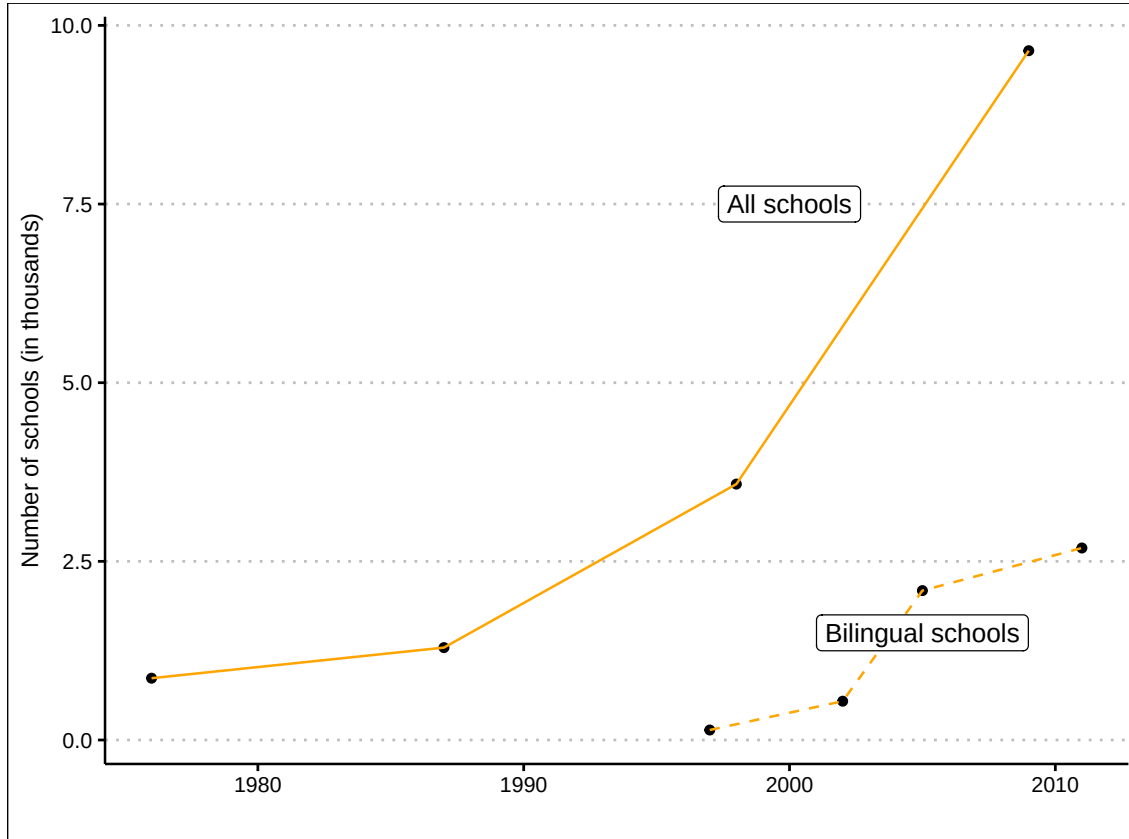
- Ni dape felen be men kumajaganya kono, i bolo kelen korata.
tile lamini dagafeeredankan hakili tije kalanden
- Ni kumamen felen be men kumajaganya kono, i bolo kelen korata.
 - O tuma, i jinena wa ?
 - Misiw be feere sugu la.
 - Ala ko tije, fo n'i ye n hakili jigin.
- Kumajaganya kumamen to fo
 - Ali, tile koratara de ! O tuma
 - A ka nogo Ala fe. N kunta
- Kumajaganya kumamen damine fo.
 - fo n'i ye n hakili jigin.
 - jon ye. N'i labennen don, an ka taa.
 - bolo. Ni wari caman t'a' kun, hali a' ka segin so.
- Tiri be dape min kora, o jaganaw lakoori
 - tile : teri, tile, ten, tile, tun, tuma, tile,
 - sanbara : sisan, sanbara, san, sugu, sanbara, somogow, sanbara
- Ja min kofolen be kumajaganya kono, o kunnafoni sehen.

tile

misi

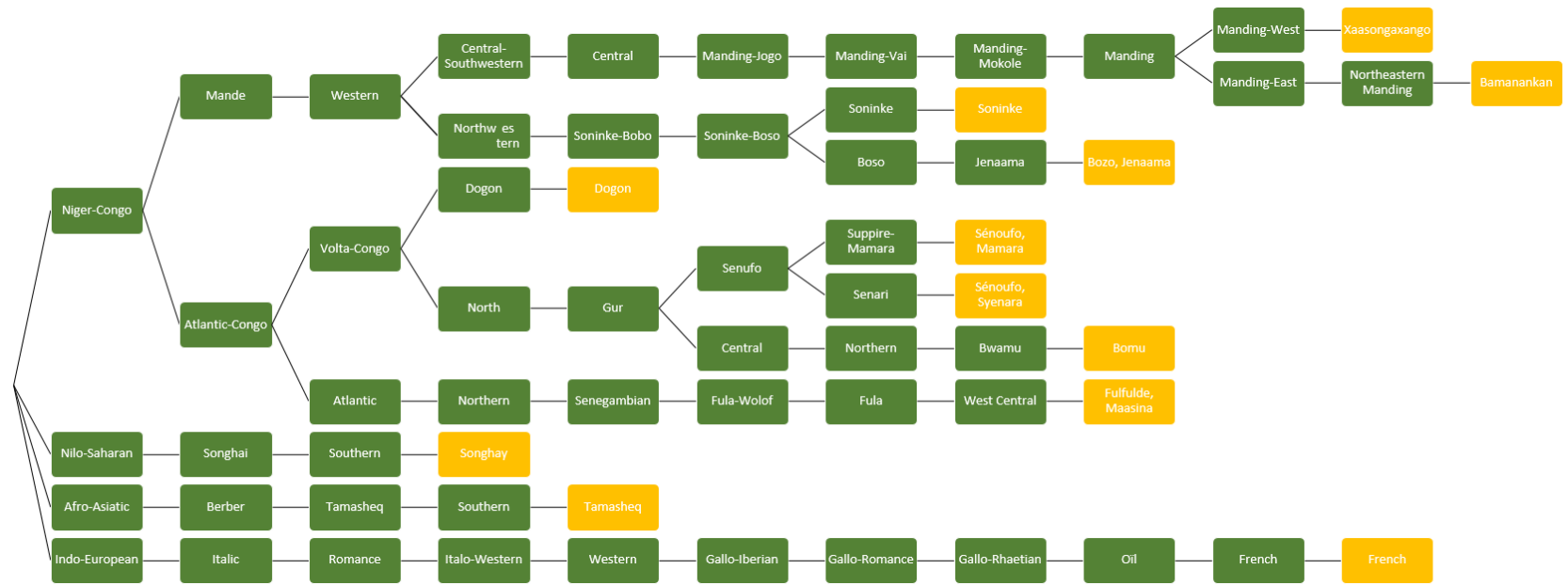
daba

Figure A.1: Examples of a textbook in Bamanankan for grade 1 students



Notes: The orange line represents the school supply, regardless of the languages of instruction used in the school. The school supply was inferred from the national census data. The orange dotted line represents the number of bilingual schools. The number of bilingual schools was retrieved from archival sources from the Bilingual Education monitoring section. The four points in time are the only years for which I have accurate data (see Table A.3).

Figure A.2: Time evolution of the number of schools



Notes: This figure presents the different language trees for the 12 languages that can be used as a language of instruction, as stated in the 1999 bilingual education reform in Mali. Ethnologue is the source used to create the trees (Eberheard, Simons, and Fennig, 2025). Yellow cells indicate languages introduced in the 1999 PRODEC reform, while green cells relate to language families. Family connections are indicated with black lines. The four language families on the left-hand side of the figure are the higher-level language families as listed in Ethnologue.

Figure A.3: Language trees for the languages of instruction in Mali

A.3 Tables

Primary education	Local language	French
Grade 1	100%	0%
Grade 2	75%	25%
Grade 3	50%	50%
Grade 4	50%	50%
Grade 5	25%	75%
Grade 6	25%	75%

Table A.1: Language repartition in the bilingual curriculum

Notes: This table comes from the work of Diarra (2020) and indicates the official time spent teaching in French and the local language at each grade of the primary cycle.

Region	Language(s)
Kayes	Soninke, Bamanankan, Fulfulde, Khassonke
Koulikoro	Bamanankan, Syenara, Mamara
Sikasso	Bamanankan, Syenara, Mamara
Segou	Bamanankan, Bomu, Mamara
Mopti	Fulfulde, Dogon, Bamanankan, Songhay, Bozo
Tombouctou	Songhay, Tamasheq, Fulfulde
Gao	Songhay, Tamasheq
Kidal	Tamasheq
Bamako	Bamanankan

Table A.2: Language choice per region

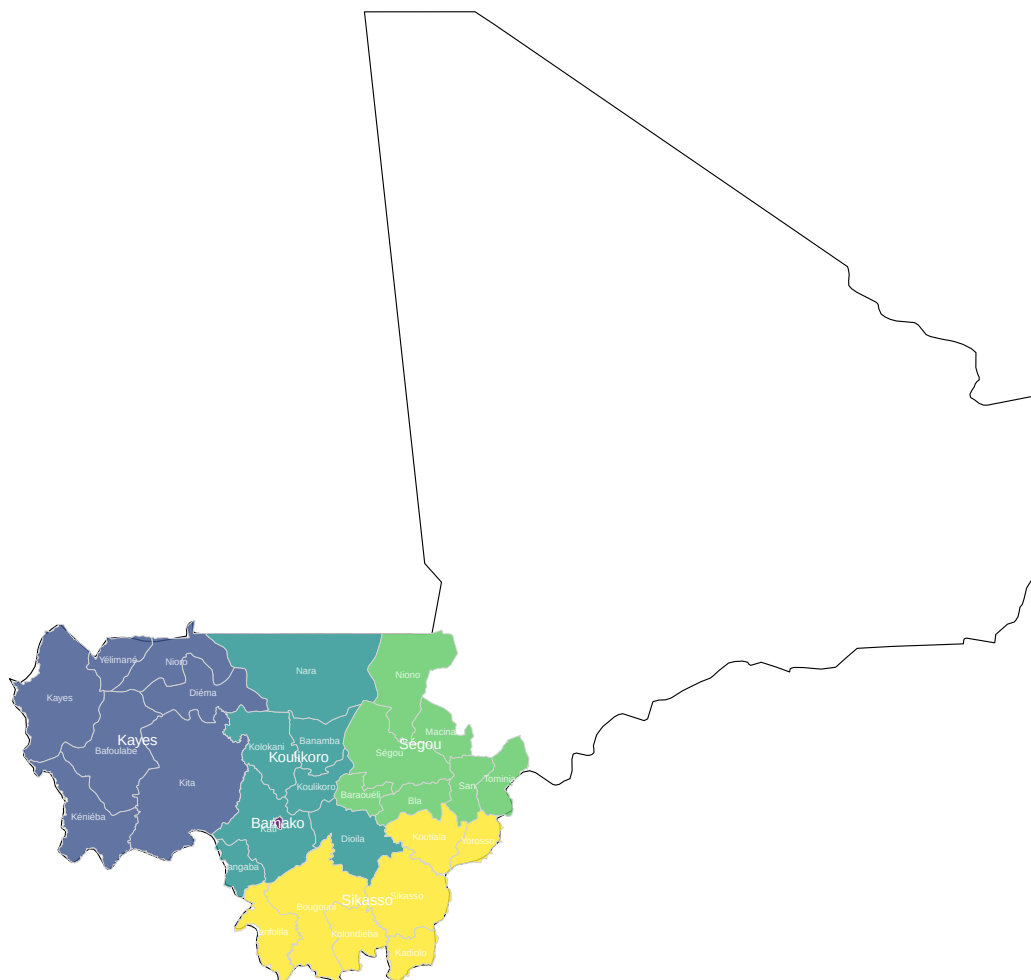
Notes: This table comes from the work of Diarra (2020) and indicates the local language(s) chosen to be languages of instruction for at least one school in each region.

Year	Geographical level	Information	Source
1994-1997	District	Number of BS per language	Trefault (1999) Diarra (2020)
2002	District	Number of BS	MEN (2003)
2005	Region	Number of BS per language	Official reports
	District	Number of BS	
2011	Commune	List of BS with info. on their status	Diarra (2013)

Table A.3: Year, geographical level, and information available for bilingual education

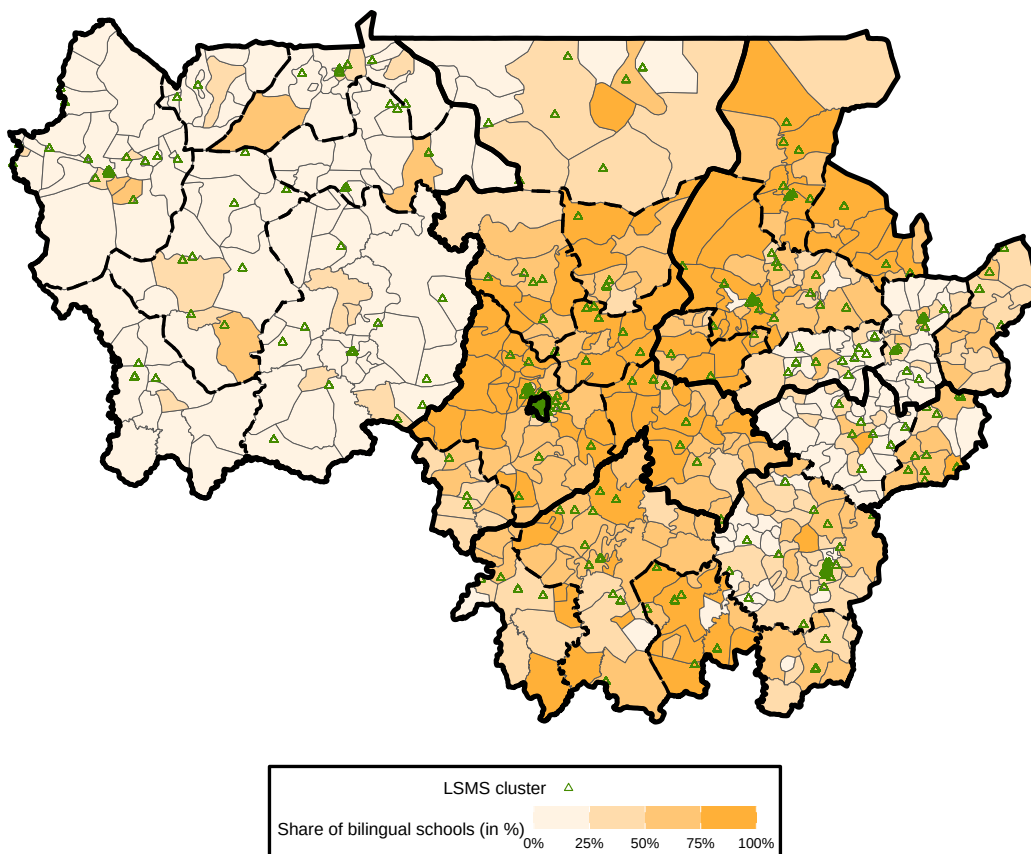
Notes: “BS” stands for bilingual schools. “1994-1997” stands for “from 1994 to 1997”. This table provides information on data retrieved regarding bilingual education supply in Mali. To document the expansion of bilingual education, I use original data from several reports published by the Bilingual Education Department within the Ministry of Education between 2002 and 2011 (Diarra, [2013](#); MEN, [2003](#)). I also use data on the first experiments from a PhD thesis in linguistics (Diarra, [2020](#)) and a field qualitative study conducted at the end of the 1990s (Trefault, [1999](#)). From the various sources detailed previously, I can document four periods: the first experiment from 1994 to 1997, followed by 2002, 2005, and 2011.

A.4 Maps



Notes: This map presents the regions and districts covered by the 2011 census on bilingual schools, and the rest of the Malian territory. Regions are indicated in colors, while districts are delimited in gray. The black lines represent the country's borders.

Figure A.4: Map of Mali with regions covered by the 2011 bilingual education census

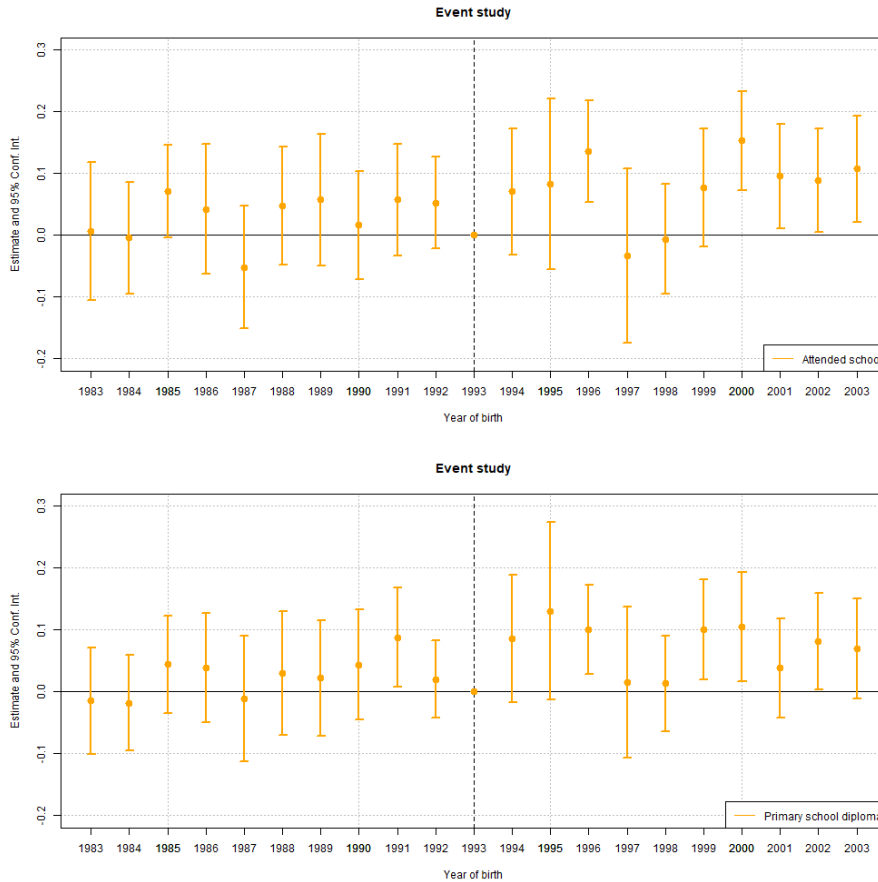


Notes: Solid black lines represent the regional borders, dashed black lines represent the district borders, and gray lines represent the communal borders. The color represents the intensity in the fraction of bilingual schools in each commune: the whiter the commune, the less the provision of bilingual education in the commune. The green triangles represent the location of LSMS clusters used in the analysis. This figure maps only the following regions considered in the empirical analysis: Bamako, Kayes, Koulikoro, Segou, and Sikasso.

Figure A.5: Bilingual education supply in 2011 and LSMS clusters by commune

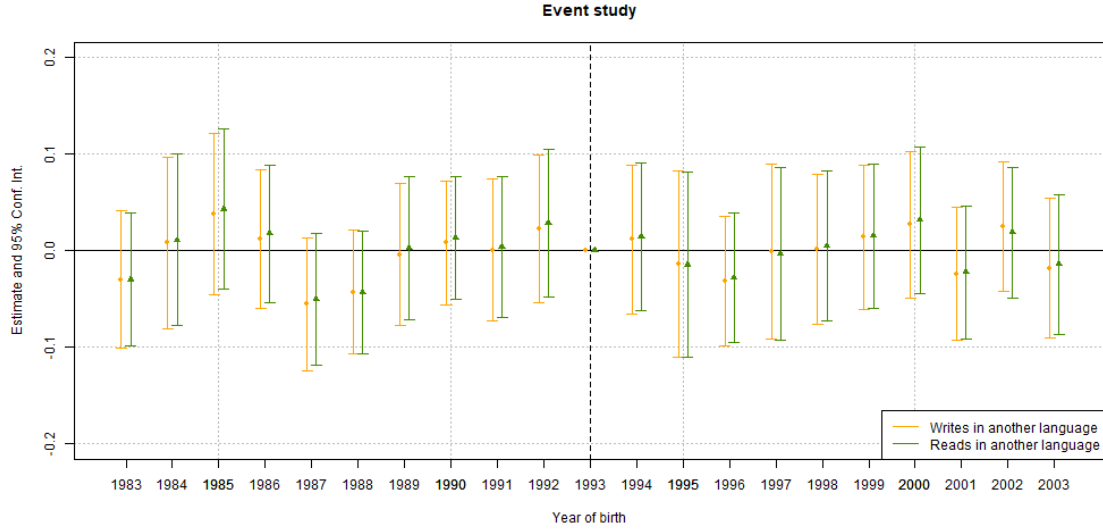
B Results

B.1 Figures



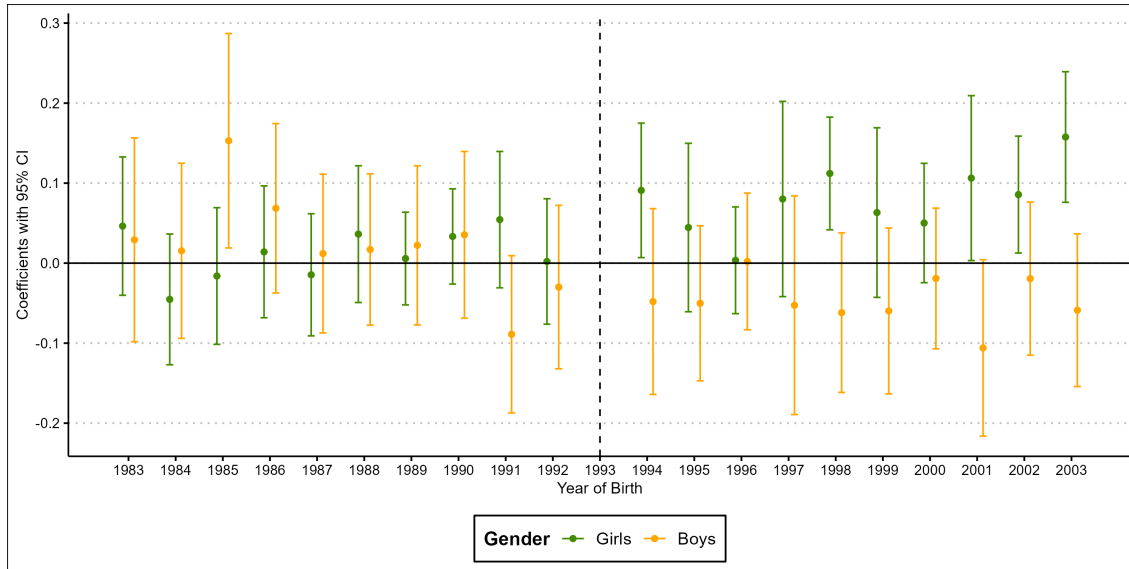
Notes: Point estimates and 95% confidence intervals are derived from an event study regression over the 1983-2003 birth cohorts, specified in Equation 2. In addition to time fixed effects, I use district fixed effects to capture that the median of bilingual education share varies across districts. To obtain these estimates, I perform two distinct regressions on two separate outcomes: (i) whether the individual attended school, and (ii) whether the individual obtained the end-of-primary-school exam, called CEP. I cluster at the LSMS cluster level, which corresponds roughly to the village level. Individuals born in 1993 are the last birth cohort to be taught entirely in French, i.e., not treated, represented by a black dashed vertical line.

Figure B.1: Event study results for schooling outcomes



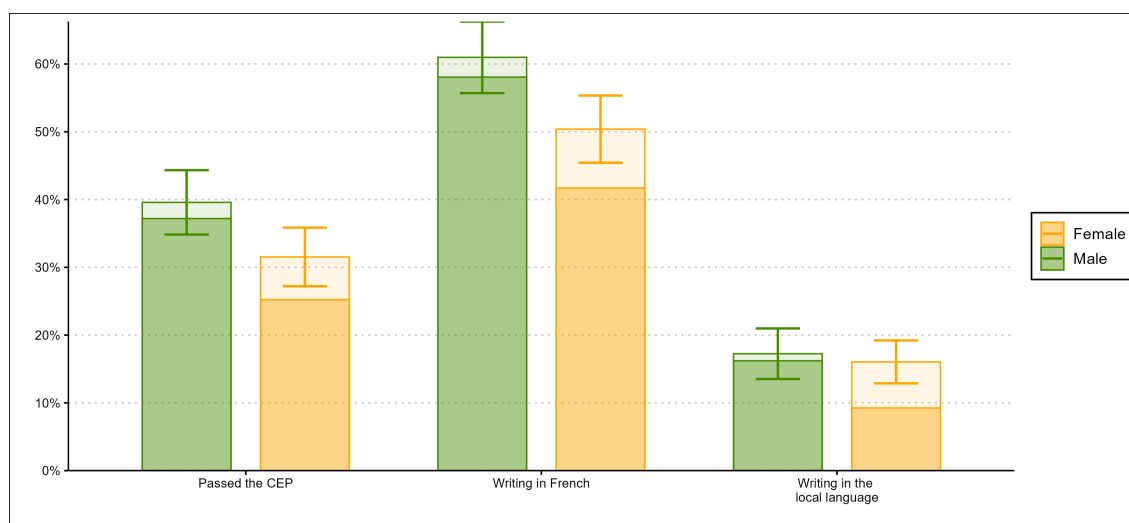
Notes: Point estimates and 95% confidence intervals are derived from an event study regression over the 1983-2003 birth cohorts, specified in Equation 2. In addition to time fixed effects, I use district fixed effects to capture variation in the median bilingual education share across districts. To obtain these estimates, I perform two distinct regressions on two separate outcomes: (i) whether the individual knows how to write in another language as the dependent variable (in orange), and (ii) whether the individual knows how to read in another language as the dependent variable (in green). I cluster the standard errors at the LSMS cluster level, which corresponds roughly to the village level. Individuals born in 1993 are the last birth cohort to be taught entirely in French (i.e., not treated), represented by a black vertical dashed line.

Figure B.2: Event study results for literacy in another language



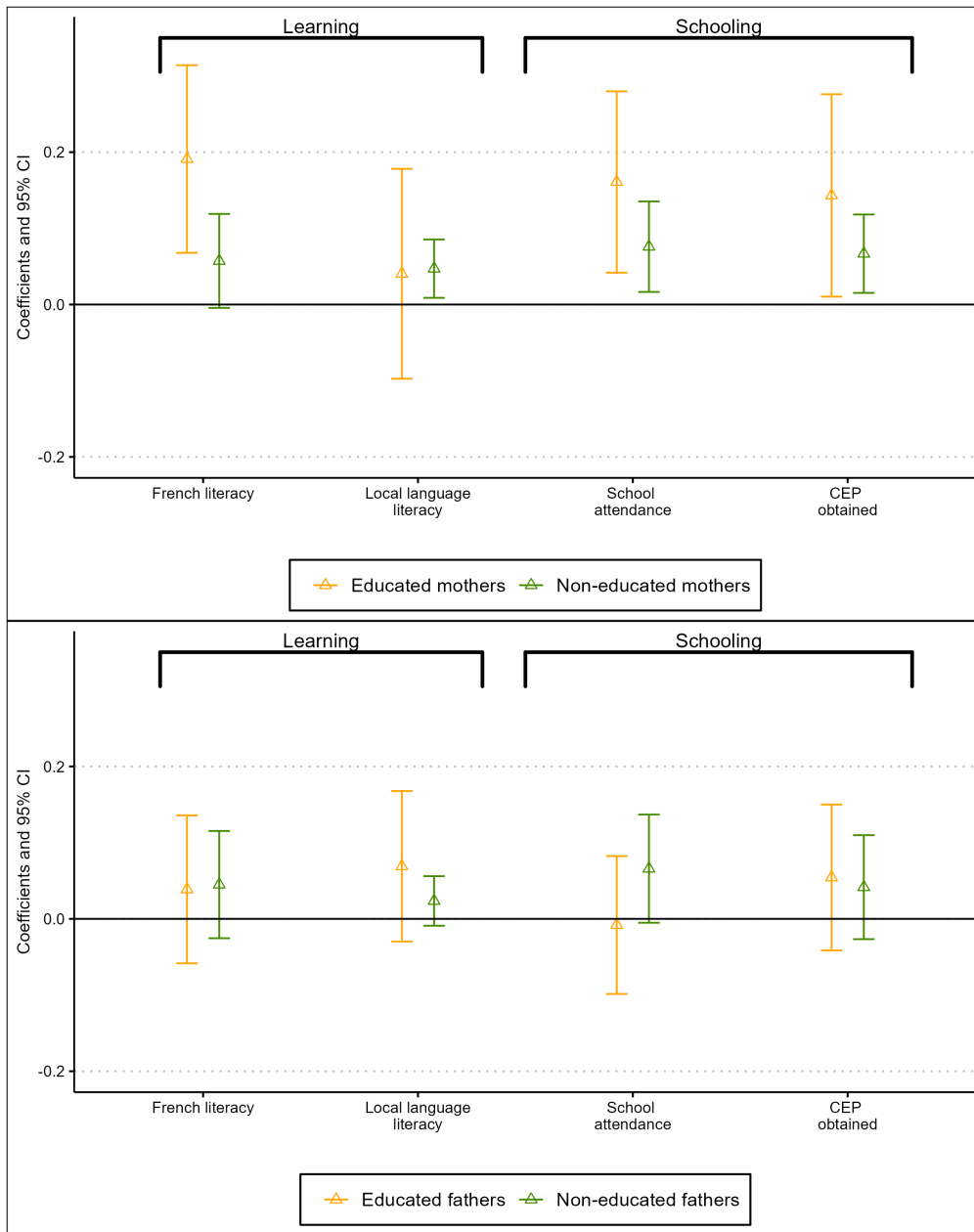
Notes: Point estimates and 95% confidence intervals are derived from an event study regression over the 1985-2003 birth cohorts, specified in Equation 2. In addition to time fixed effects, I use district fixed effects to capture variation in the median bilingual education share across districts. The outcome considered here is whether the individual knows how to read and write in the local language. I interact the α_t and β_t coefficients with a dummy indicating the respondent's gender. I cluster the standard errors at the LSMS cluster level, which corresponds roughly to the village level. Individuals born in 1993 are the last birth cohort to be taught entirely in French (i.e., not treated), represented by a black vertical dashed line.

Figure B.3: Event study results for literacy in the local language, by gender



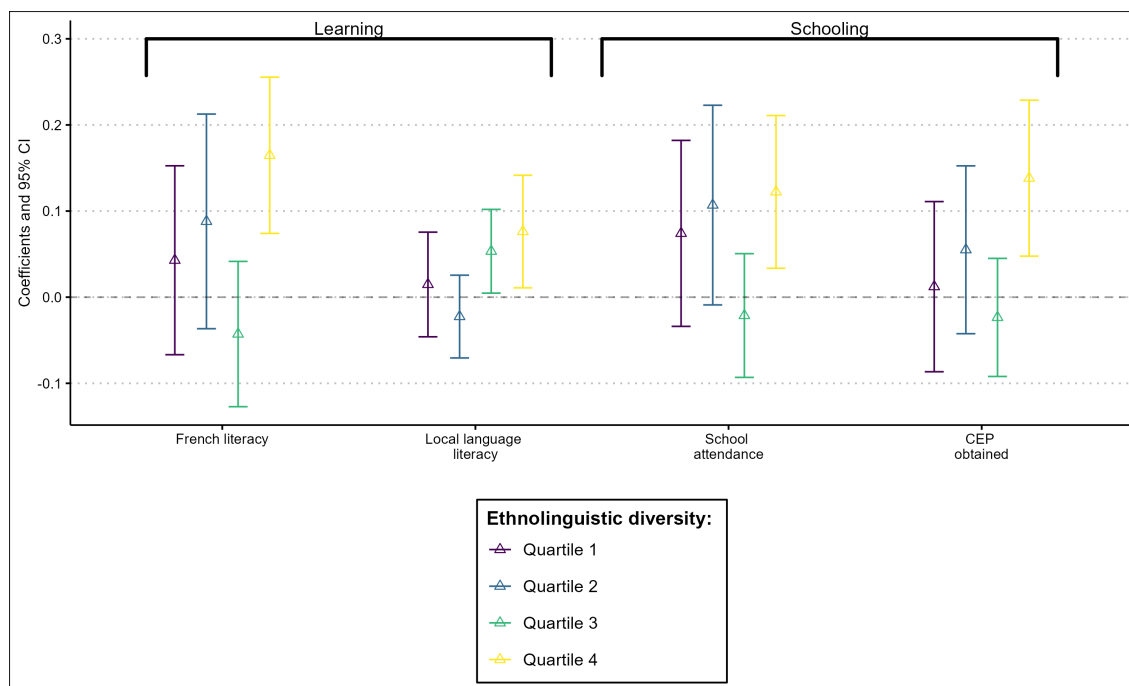
Notes: The solid bars represent the outcome mean at baseline, for the three outcomes described on the x-axis. The shadowed areas on top of the solid bars illustrate the effect of bilingual education on each of these outcomes, with the corresponding 95% confidence intervals. Each of these results is depicted at the gender level. Point estimates and 95% confidence intervals are derived from a TWFE linear regression, with the interaction dummy interacted with a gender dummy.

Figure B.4: TWFE coefficient estimates of bilingual education effects relative to the outcome means, by gender



Notes: Point estimates and 95% confidence intervals are derived from a TWFE linear regression on several outcomes: (1) and (2) are literacy measures (i.e., ability to write and read) in French and in the local language, and (3) (4) and (5) are school achievements indicators. I use district fixed effects to capture the fact that the median of bilingual education share varies across districts. I cluster at the LSMS cluster level, which corresponds roughly to the commune level. To obtain heterogeneous effects, I interact the DiD coefficient in Equation 1 with a dummy variable equal to one if the mother or father attended school, and I also add this dummy variable to the set of fixed effects.

Figure B.5: TWFE coefficient estimates of bilingual education effects, by parental education



Notes: Point estimates and 95% confidence intervals are derived from a TWFE linear regression on several outcomes: (1) and (2) are literacy measures (i.e., ability to write and read) in French and in the local language, and (3) (4) and (5) are school achievements indicators. I use district fixed effects to capture the fact that the median of bilingual education share varies across districts. I cluster at the LSMS cluster level, which corresponds roughly to the commune level. To obtain heterogeneous effects, I interact the DiD coefficient in Equation 1 with quartiles of the ethnolinguistic diversity index, and add these quartiles to the set of fixed-effects as well.

Figure B.6: TWFE coefficient estimates of bilingual education effects, by linguistic diversity

B.2 Tables

	2 to 4 yo: The child knows at least...		5 to 6 yo: The child works in...		
Dependent Variables:	10 letters in the alphabet	How to read 4 simple words	Farming activity	A firm	Selling activity
Model:	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
The child is a girl	-0.012* (0.007)	-0.011 (0.009)	-0.061** (0.024)	-0.0005 (0.019)	0.012 (0.010)
Controls:	Y	Y	Y	Y	Y
<i>Fixed-effects</i>					
Region FE	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	6,423	6,423	1,587	1,589	1,588

Clustered (MICS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression, using MICS data. I control for age, weight-for-age z-score and the household wealth score for results presented in (1) and (2). I control for age and the household wealth score for results presented in (3) and (4). The level of analysis is at the individual level, i.e., one observation is one person. The differences in The number of observations is related to the different samples considered between Columns (1) and (2) on one side and Columns (3) and (4) on the other side. I use the weights provided in the MICS data and cluster the standard errors at the enumeration area level.

Table B.1: Regression analysis on prio exposure to French for boys

Dependent Variables:	The husband/partner has the final say on...				
	Women health	Large household	Small household	Visiting people outside	Food cooked
		purchase	purchase	of the household	
Model:	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
Has a child born after 1994	0.080** (0.040)	0.053 (0.042)	0.023 (0.046)	0.089** (0.039)	0.036 (0.032)
High exposed commune	0.018 (0.056)	0.011 (0.058)	-0.006 (0.064)	0.069 (0.059)	0.055 (0.046)
Has a child born after 1994 × High exposed commune	-0.042 (0.056)	-0.027 (0.057)	-0.010 (0.059)	-0.121** (0.056)	-0.044 (0.047)
<i>Fixed-effects</i>					
District FE	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Mean of Y	0.706	0.658	0.568	0.544	0.271
Observations	7,513	7,513	7,505	7,510	7,511
R ²	0.04493	0.04304	0.05220	0.08123	0.09514

Clustered (cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw DHS data. I use the weights given in the DHS data, and cluster the standard errors at the enumeration area. Clustering at the treatment level, i.e. the commune, produces the same results.

Table B.2: TWFE results on decision-making outcomes

Dependent Variables:	French literacy	Local language literacy	Attended school	Primary school diploma
Model:	(1)	(2)	(3)	(4)
<i>Panel 1 – Heterogeneity variable: Bambara as the main language of the location</i>				
HV	0.163*** (0.060)	0.032 (0.023)	0.121** (0.052)	0.127** (0.053)
DiD	0.106* (0.064)	0.037 (0.039)	0.109* (0.058)	0.058 (0.052)
DiD x HV	-0.070 (0.069)	0.001 (0.040)	-0.066 (0.061)	-0.024 (0.056)
<i>Panel 2 – Heterogeneity variable: Urban location</i>				
HV	0.326*** (0.040)	0.055*** (0.020)	0.264*** (0.035)	0.275*** (0.037)
DiD	0.054 (0.033)	0.017 (0.020)	0.058* (0.032)	0.025 (0.029)
DiD x HV	-0.028 (0.042)	0.058** (0.029)	-0.022 (0.042)	0.024 (0.037)
<i>Panel 3 – Heterogeneity variable: Location less than 10km from a road (in 2000)</i>				
HV	0.152*** (0.026)	0.027* (0.015)	0.147*** (0.026)	0.115*** (0.021)
DiD	0.038 (0.042)	0.018 (0.022)	0.042 (0.037)	0.016 (0.034)
DiD x HV	0.043 (0.060)	0.046* (0.026)	0.044 (0.050)	0.060 (0.052)
<i>Panel 4 – Heterogeneity variable: All BS in the commune are still bilingual in 2011</i>				
HV	-0.116** (0.058)	-0.009 (0.018)	-0.102** (0.045)	-0.073 (0.052)
DiD	-0.024 (0.035)	0.035 (0.022)	-0.019 (0.034)	-0.007 (0.029)
DiD x HV	0.135*** (0.045)	0.007 (0.028)	0.137*** (0.043)	0.083** (0.038)
<i>Fixed-effects</i>				
District FE	Yes	Yes	Yes	Yes

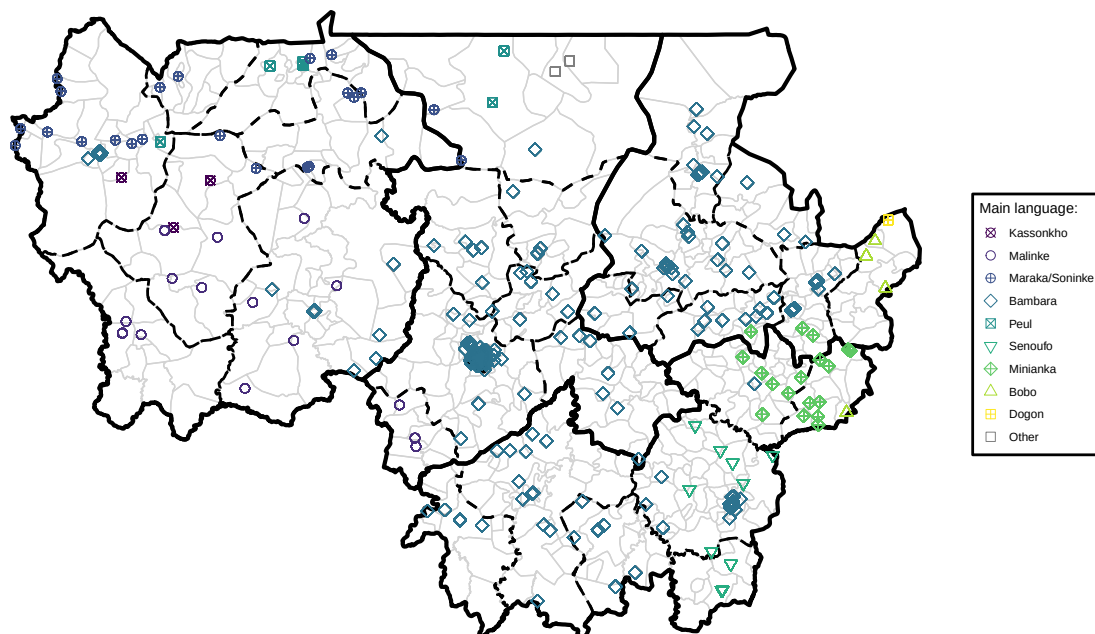
Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: “BS” stands for bilingual schools. All dependent variables are binary outcomes. The model estimated is a linear regression. Literacy in one language is equal to one when the individual knows how to write and read in this language. The level of analysis is at the individual level, i.e., one observation is one person. I use the weights provided in the LSMS data and cluster the standard errors at the enumeration area level. Clustering at the treatment level, i.e., the commune, produces the same results. HV stands for heterogeneity variable. DiD stands for the difference-in-difference interaction coefficient. I report only this coefficient, along with the heterogeneity variable coefficient, detailed in the first column, and the interaction coefficient between the DiD dummy and the heterogeneity dummy.

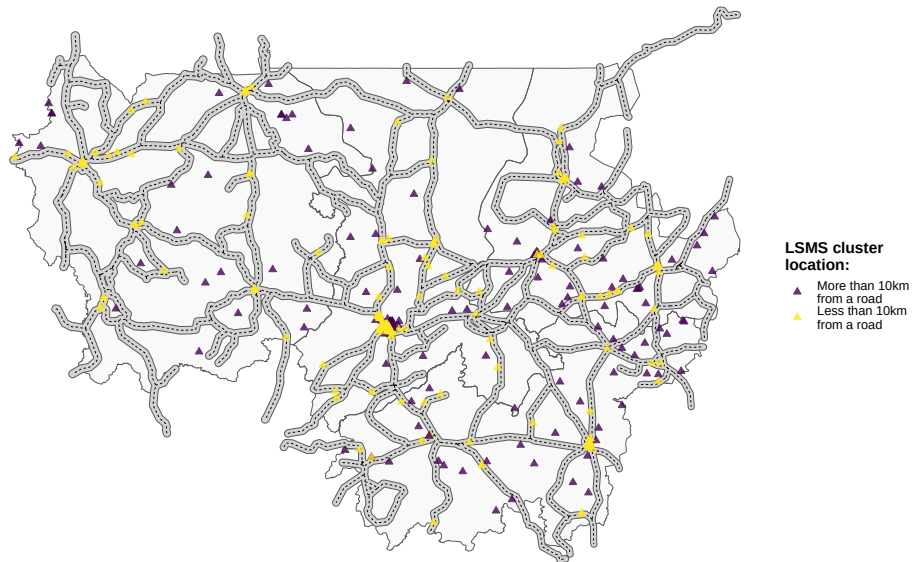
Table B.3: Heterogeneity results on language, urban, road access, and continuity in the bilingual education provision

B.3 Maps



Notes: Solid black lines represent the regional borders, dashed black lines represent the district borders, and gray lines represent the communal borders. The dots are the LSMS cluster locations. Colors and shapes indicate the dominant language of the commune in which these clusters are located, as given in the 2009 census.

Figure B.7: Map of LSMS clusters with the main language spoken

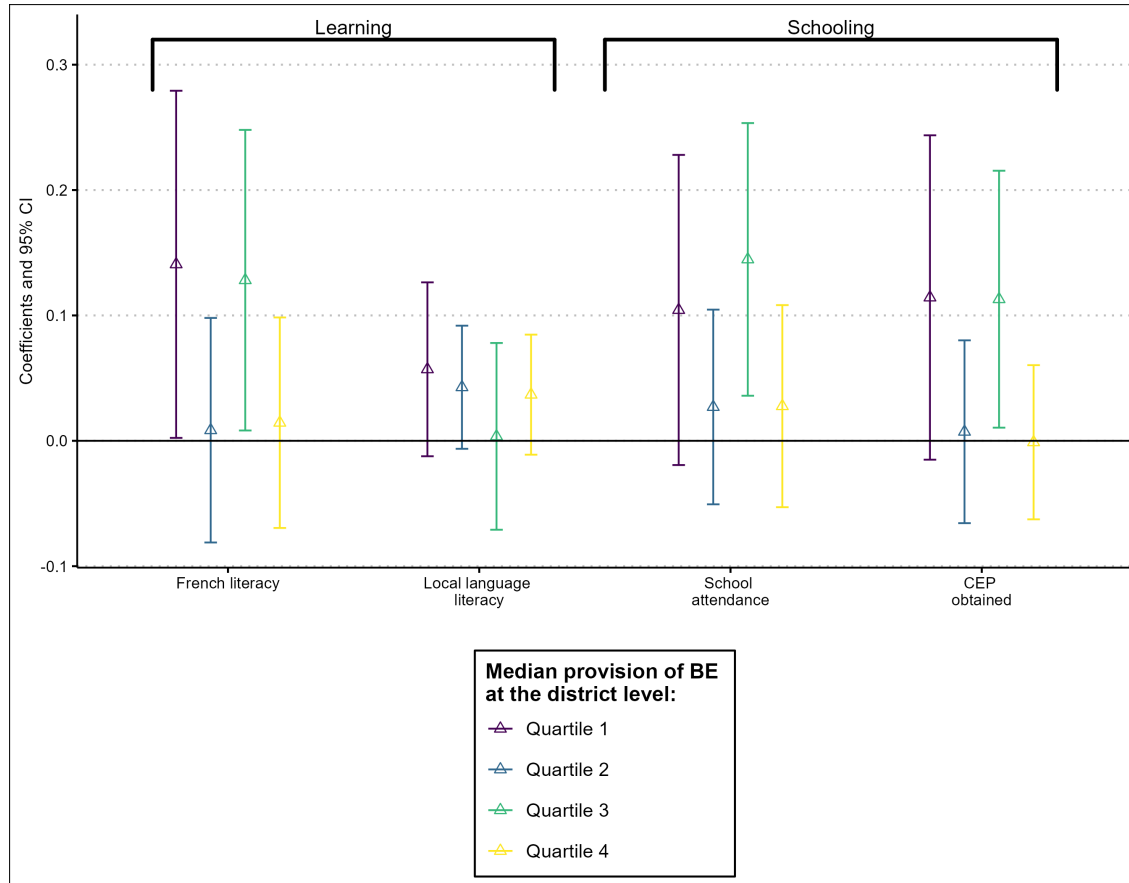


Notes: Solid black lines represent the regional borders. Roads represented on the map are taken from the Michelin map from 2000 (Müller-Crepon, 2023; Müller-Crepon, Hunziker, and Cederman, 2021). The dots represent the LSMS cluster locations, and their colors vary according to whether they are located within 10 kilometers (as the crow flies) of the closest road.

Figure B.8: Map of road access to LSMS clusters

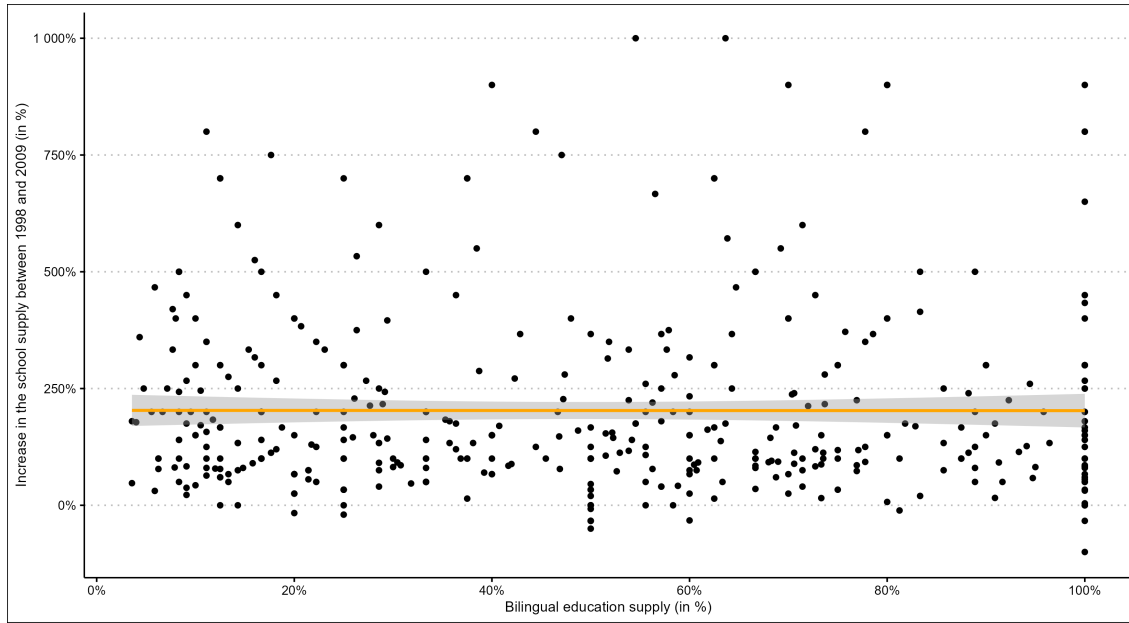
C Robustness checks

C.1 Figures



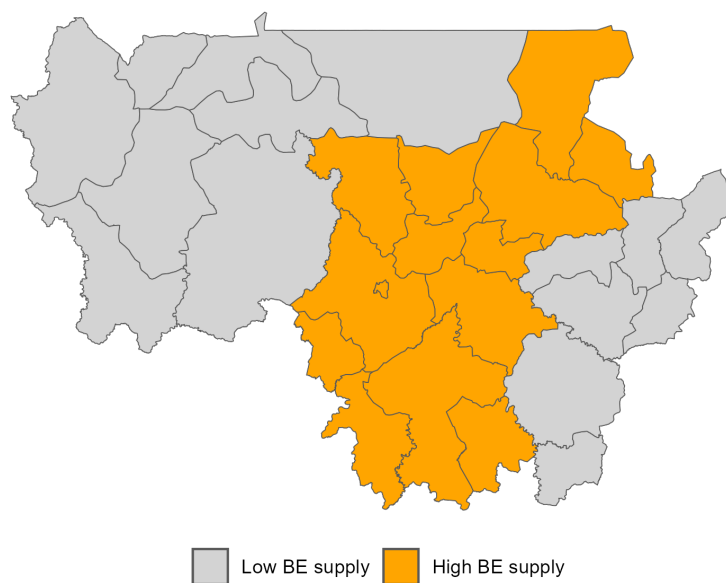
Notes: Point estimates and 95% confidence intervals are derived from a TWFE linear regression on several outcomes: (1) and (2) are literacy measures (i.e., ability to write and read) in French and in the local language, and (3) (4) and (5) are school achievements indicators. I use district fixed effects to capture the fact that the median of bilingual education share varies across districts. I cluster at the LSMS cluster level, which corresponds roughly to the commune level. To obtain heterogeneous effects, I interact the DiD coefficient in Equation 1 with quartiles of bilingual education share, and add these quartiles to the set of fixed-effects as well.

Figure C.1: TWFE coefficient estimates of bilingual education effects, by the median share of bilingual education provision



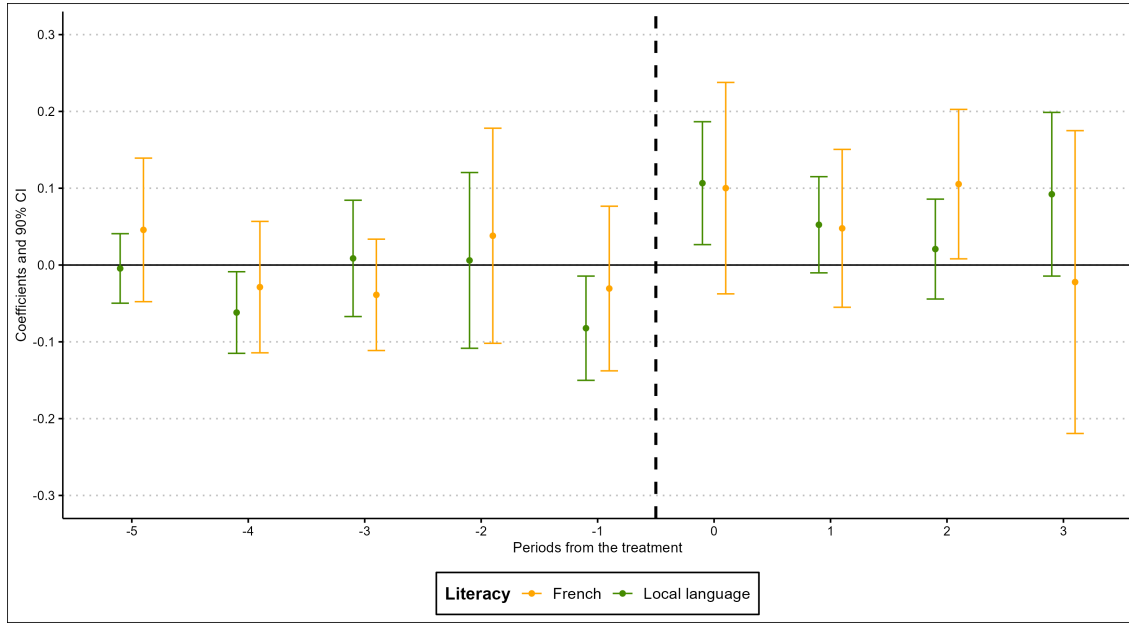
Notes: One black dot corresponds to one commune in the sample. The y-axis depicts the increase in the number of schools from 1998 to 2009 as given in the census, in percentage. The x-axis represents the share of bilingual schools among the total number of schools within the commune. The orange line is obtained from regressing the share of bilingual schools on the increase in schools. The shadowed area represents the 95% confidence interval of the estimated regression coefficient.

Figure C.2: Correlation between school building and bilingual education supply



Notes: BE stands for Bilingual Education. The districts highlighted in orange are districts with a relatively high share of bilingual schools in the total school supply. These orange districts are used for performing the analysis described in Section [5.4](#).

Figure C.3: Map of districts considered in the analysis described in Section [5.4](#)



Notes: Point estimates and 90% confidence intervals are derived from a staggered analysis using Borusyak, Jaravel, and Spiess (2024). The dotted black vertical line indicates the last pre-treatment period. The two sets of points and coefficient estimates are obtained through two separate regressions. The treatment is estimated at the district level. I use region-fixed effects, and I cluster at the LSMS cluster level, which corresponds roughly to the village level.

Figure C.4: Staggered analysis on French and local language literacy

C.2 Tables

	Local language		French		Attended	Primary school
Dependent Variables:	Writing	Reading	Writing	Reading	school	diploma
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Born after 1994	0.017 (0.013)	0.015 (0.012)	0.142*** (0.020)	0.140*** (0.020)	0.160*** (0.019)	0.076*** (0.017)
High exposed commune	-0.0004 (0.015)	-0.0007 (0.014)	-0.003 (0.037)	-0.004 (0.036)	-0.012 (0.032)	-0.006 (0.031)
Born after 1994 $\times$ High exposed commune	0.019 (0.019)	0.022 (0.018)	0.050* (0.030)	0.051* (0.030)	0.058** (0.028)	0.039 (0.026)
<i>Fixed-effects</i>						
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Mean of Y	0.123	0.118	0.489	0.484	0.495	0.305
Observations	8,636	8,623	8,636	8,633	8,635	8,635
R ²	0.00695	0.00671	0.11743	0.11475	0.08517	0.09628

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw LSMS data. I use region fixed-effects, as the treatment uses the regional median as a reference point. I use the weights given in the LSMS data, and cluster the standard errors at the enumeration area. Clustering at the treatment level, i.e. the commune, produces the same results.

Table C.1: TWFE results with region fixed-effects

	French	Local language	Attended	Primary school	French	Local language	Attended	Primary school
Dependent Variables:	literacy	literacy	school	diploma	literacy	literacy	school	diploma
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Variables</i>								
Born after 1994	0.206*** (0.028)	0.019 (0.020)	0.194*** (0.039)	0.089*** (0.021)	0.110*** (0.024)	6.54×10^{-5} (0.014)	0.142*** (0.021)	0.066*** (0.021)
High exposed commune	0.079** (0.038)	0.009 (0.028)	0.076 (0.050)	0.076** (0.035)	0.015 (0.063)	0.039* (0.022)	-0.007 (0.049)	0.006 (0.053)
Born after 1994 × High exposed commune	0.011 (0.040)	0.022 (0.029)	0.020 (0.047)	0.003 (0.032)	0.060 (0.038)	0.046** (0.022)	0.076** (0.034)	0.065* (0.034)
Sub-sample	Close	Close	Close	Close	Far	Far	Far	Far
<i>Fixed-effects</i>								
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>								
Observations	3,064	3,064	3,064	3,064	5,572	5,572	5,571	5,571

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. Literacy in one language is equal to one when the individual knows how to write and read in this language. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw LSMS data. I use the weights given in the LSMS data, and cluster the standard errors at the enumeration area. Clustering at the treatment level, i.e. the commune, produces the same results. I define the sub-sample of commune far (close) to the the median when it has a bilingual education supply at least (below) 9 percentage points higher or lower than the median. 9pp corresponds to the median distance to the median. I use district fixed effects in all regressions.

Table C.2: TWFE results, by distance to the district median

Dependent Variables:	French literacy	Local language literacy	Attended school	Primary school diploma
Model:	(1)	(2)	(3)	(4)
<i>Panel 1 – Heterogeneity variable: High school construction rate in the location</i>				
HV	-0.081** (0.032)	-0.015 (0.019)	-0.099*** (0.033)	-0.062** (0.025)
DiD	0.109*** (0.035)	0.063*** (0.022)	0.091*** (0.034)	0.082** (0.032)
DiD x HV	-0.183*** (0.048)	-0.086*** (0.027)	-0.148*** (0.049)	-0.126*** (0.041)
<i>Panel 2 – Heterogeneity variable: Respondent born in the commune</i>				
HV	0.013 (0.018)	-0.034*** (0.012)	-0.013 (0.020)	-0.004 (0.018)
DiD	-0.012 (0.040)	-0.001 (0.026)	-0.030 (0.038)	0.002 (0.035)
DiD x HV	0.084** (0.041)	0.052* (0.028)	0.113*** (0.041)	0.050 (0.038)
<i>Panel 3 – Heterogeneity variable: Attended school</i>				
HV	0.774*** (0.016)	0.164*** (0.015)		
DiD	-0.011 (0.019)	0.0007 (0.015)		
DiD x HV	0.034 (0.024)	0.050** (0.025)		
<i>Fixed-effects</i>				
District FE	Yes	Yes	Yes	Yes

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e., one observation is one person. I use the weights provided in the LSMS data and cluster the standard errors at the enumeration area level. Clustering at the treatment level, i.e., the commune, produces the same results. “HV” stands for heterogeneity variable. “DiD” stands for the difference-in-difference interaction coefficient. I report only this coefficient, along with the heterogeneity variable coefficient, detailed in the first column, and the interaction coefficient between the DiD dummy and the heterogeneity dummy.

Table C.3: Heterogeneity results on school construction and migration

Dependent Variable:	Number of schools in the commune
Model:	(1)
<i>Variables</i>	
High exposed commune	-0.418 (0.755)
2009 census	13.9*** (3.42)
High exposed commune $\times$ 2009 census	-5.59* (2.77)
<i>Fixed-effects</i>	
District FE	Yes
Mean of Y	11.56
<i>Fit statistics</i>	
Observations	918
R ²	0.31844

Clustered (District FE) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The model estimated is a linear regression. The level of analysis is at the year-commune level, i.e., one observation is one commune at a given year. All data on the number of schools is taken from the 1998 and 2009 censuses. “2009 census” is a dummy equal to one when the number of schools is taken from the 2009 census, and zero when it comes from the 1998 census. It is equivalent to the dummy “Born before 1994” in Equation 1, as the number of schools in 1998 was before the 1999 PRODEC reform.

Table C.4: TWFE results on the number of schools

	Local language literacy	French literacy	Attended school	Primary school diploma
Model:	(1)	(2)	(3)	(4)
<i>Panel 1 – Controlling for the number of school-age children per school</i>				
Born after 1994	0.136*** (0.020)	0.006 (0.011)	0.156*** (0.019)	0.072*** (0.016)
High exposed commune	0.019 (0.041)	0.014 (0.015)	0.012 (0.034)	0.016 (0.034)
DiD	0.054* (0.028)	0.039** (0.018)	0.059** (0.026)	0.041* (0.025)
Average number of school-age children per school	0.001 (0.0010)	0.0009** (0.0004)	0.001 (0.0009)	0.001 (0.0008)
<i>Panel 2 – Controlling for the number of school-age children per school</i>				
Born after 1994	0.144*** (0.019)	0.008 (0.011)	0.163*** (0.018)	0.079*** (0.016)
High exposed commune	0.092*** (0.028)	0.036** (0.015)	0.080*** (0.028)	0.077*** (0.022)
DiD	0.047* (0.028)	0.037** (0.017)	0.052** (0.026)	0.035 (0.024)
Number of schools in the commune	0.002*** (0.0003)	0.0004** (0.0002)	0.002*** (0.0002)	0.001*** (0.0003)
<i>Panel 3 – Adding linguistic areas fixed-effects</i>				
Born after 1994	0.006 (0.011)	0.136*** (0.020)	0.155*** (0.019)	0.074*** (0.016)
High exposed commune	0.021 (0.016)	0.018 (0.040)	0.019 (0.034)	0.015 (0.034)
DiD	0.039** (0.017)	0.054* (0.028)	0.060** (0.027)	0.040 (0.024)
Linguistic area FE	Y	Y	Y	Y

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: “DiD” stands for the coefficient associated with the interaction between high exposed commune and being born after 1994. All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. I use the weights given in the LSMS data, use fixed-effects at the district level, and cluster the standard errors at the enumeration area. Literacy in one language is equal to one when the individual knows how to write and read in this language. The average number of school-age children per school is obtained by taking the ratio of the total number of children between 7 and 12 in 2009 (census) divided by the number of schools. I define linguistic areas by using the main language spoken in the commune in the census.

Table C.5: TWFE results with additional controls

	French	Local language	Attended school	Primary school
Dependent Variables:	literacy	literacy	literacy	diploma
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Born after 1994	0.093***	0.001	0.129***	0.059**
	(0.026)	(0.018)	(0.024)	(0.024)
High exposed commune	-0.045	0.022	-0.048	-0.024
	(0.050)	(0.026)	(0.046)	(0.038)
Born after 1994	0.071**	0.006	0.059*	0.028
× High exposed commune	(0.034)	(0.022)	(0.032)	(0.030)
<i>Fixed-effects</i>				
District FE	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,104	5,104	5,104	5,104

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. Literacy in one language is equal to one when the individual knows how to write and read in this language. The level of analysis is at the individual level, i.e., one observation is one person. The differences in the number of observations are related to missingness of the outcome of interest in the raw LSMS data. I use the weights provided in the LSMS data and cluster the standard errors at the enumeration area level. Clustering at the treatment level, i.e., the commune, produces the same results. I define a commune as highly exposed to bilingual education when the number of bilingual schools in this location exceeds half of the total school supply. The sample is limited to districts highlighted in Figure C.3.

Table C.6: TWFE results using a 50% bilingual education supply cutoff

	Local language	French	Attended	Primary school
Dependent Variables:	literacy	literacy	school	diploma
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Born after 1994	0.174*** (0.029)	0.011 (0.015)	0.199*** (0.029)	0.089*** (0.021)
High exposed commune	0.160*** (0.037)	0.046** (0.020)	0.148*** (0.037)	0.128*** (0.027)
Born after 1994 $\times$ High exposed commune	0.063 (0.040)	0.033 (0.025)	0.046 (0.040)	0.035 (0.036)
<i>Fixed-effects</i>				
District FE	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Mean of Y	0.434	0.112	0.460	0.260
Observations	4,183	4,183	4,183	4,183
R ²	0.16701	0.07334	0.15884	0.13867

Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: All dependent variables are binary outcomes. The model estimated is a linear regression. The level of analysis is at the individual level, i.e. one observation is one person. The differences in the number of observations is related to missingness of the outcome of interest in the raw LSMS data. I use district fixed-effects. I use the weights given in the LSMS data, and cluster the standard errors at the enumeration area. Clustering at the treatment level, i.e. the commune, produces the same results. The sample is restricted to clusters located at least 2km from the borders of the communes.

Table C.7: TWFE results restricting the sample to communes far from the communal borders

Dependent Variables:	Born here (0/1)	Age at the school entry
Model:	(1)	(2)
<i>Variables</i>		
High exposed commune	0.004 (0.003)	-0.003 (0.041)
<i>Fixed-effects</i>		
District FE	Yes	Yes
<i>Fit statistics</i>		
Mean of Y	0.989	6.60
Observations	7,727	4,121
R ²	0.00713	0.08627

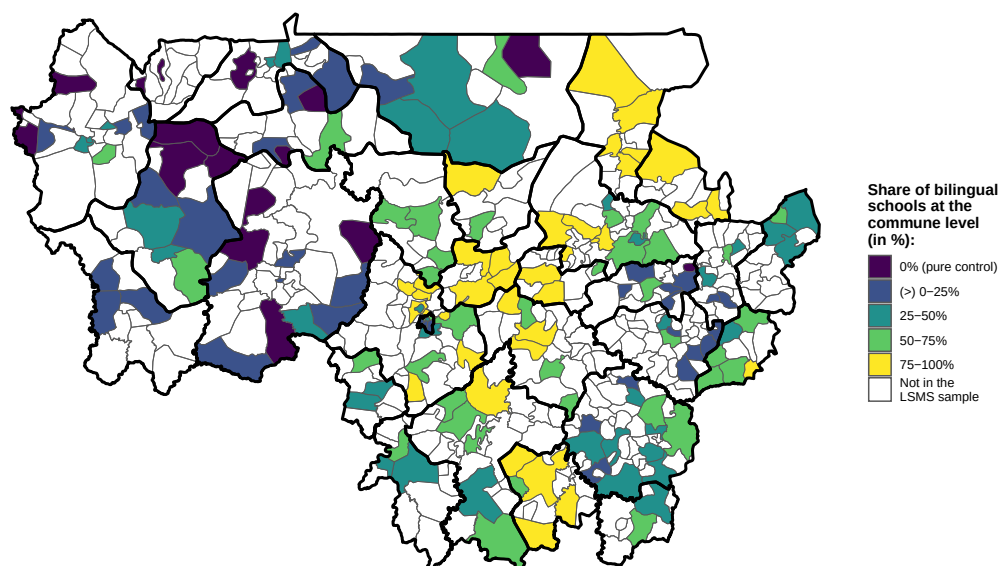
Clustered (LSMS cluster) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: I use a linear regression specification to obtain the coefficient estimates. I use district fixed effects, and I cluster at the LSMS cluster level, which corresponds roughly to the village level, and use household weights given in the LSMS survey. The mean of the outcomes is a weighted mean, using the same weights. The dependent variables are in Column (1) whether the individual was born in the surveyed village and in Column (2) the age when the individual started school for individuals who went to school (explaining the decrease in the number of observations).

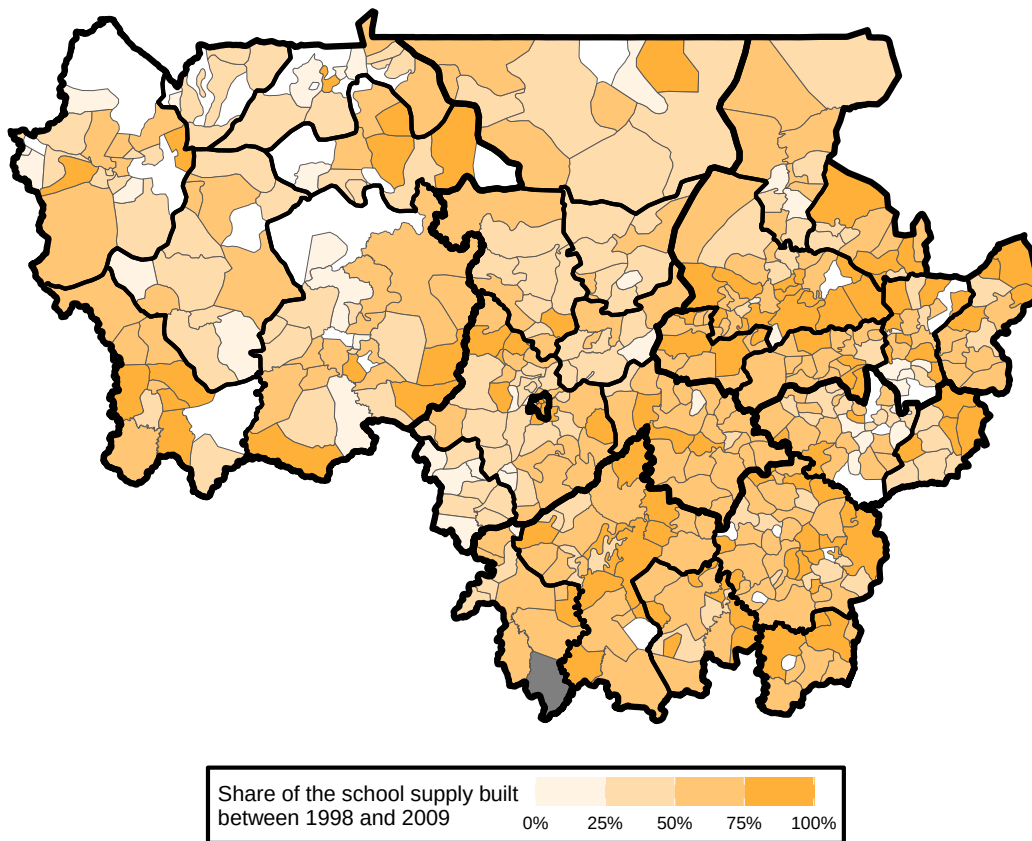
Table C.8: Correlation between treatment and potential confounders

C.3 Maps



Notes: Solid black lines represent the district borders, and gray lines the communal ones. Colors indicate the fraction of the bilingual education supply for communes in the LSMS sample.

Figure C.5: Map of binned bilingual education supply



Notes: Solid black lines represent the regional borders, dashed black lines represent the district borders, and gray lines represent the communal borders. The color represents the growth rate of the education supply at the commune level: the whiter the commune, the fewer schools were built between 1998 and 2009, in comparison to the existing school supply in 1998.

Figure C.6: Map of the education growth rate between 1998 and 2009 at the commune level

D Cost effectiveness analysis

Finally, I benchmark the estimated cost per student of bilingual education at scale in the case of Mali. To do so, I use World Bank reports to assess the implementation costs and get information on the target population (Bank, 2000). The total costs cover production and distribution of textbooks and teaching materials, the training of teachers, school principals, inspectors, and educational advisors, and the evaluation of students' and teachers' academic performance (\$3.55 million). This cost is estimated for 400 bilingual schools. From the 1998 census, I infer that one-fourth of the primary school supply is affected by the reform at its onset in the early 2000s, representing roughly 350,000 students. To do so, I use the number of children between 7 and 12 from 2000 to 2002, as they are explicitly the targeted population in the report, i.e., individuals born between 1988 and 1995 in the 2009 census. The estimated cost per targeted student is then \$10.6, well below the average cost of \$17.97 of the most effective interventions in SSA, in the recent review done by Angrist et al. (2023).